Project - 1: Deploying a Multi-Tier Website Using AWS EC2

Description:

Amazon Elastic Compute Cloud (Amazon EC2) provides scalable computing capacity in the Amazon Web Services (AWS) cloud. Using Amazon EC2 eliminates your need to invest in hardware up front so you can develop and deploy applications faster. You can use Amazon EC2 to launch as many or as few virtual servers as you need, configure security and networking, and manage storage. Amazon EC2 enables you to scale up or down to handle changes in requirements or spikes in popularity, reducing your need to forecast traffic

Problem Statement:

Company ABC wants to move their product to AWS. They have the following things set up right now:

1. MySQL DB
2. Website (PHP)

The company wants high availability on this product, therefore wants Auto Scaling to be enabled on this website

Steps To Solve:

1. Launch an EC2 Instance

2. Enable Auto Scaling on these instances (minimum 2)

3. Create an RDS Instance

4. Create Database & Table in RDS instance:

a. Database name: intel

b. Table name: data

c. Database password: intel123

5. Change hostname in website

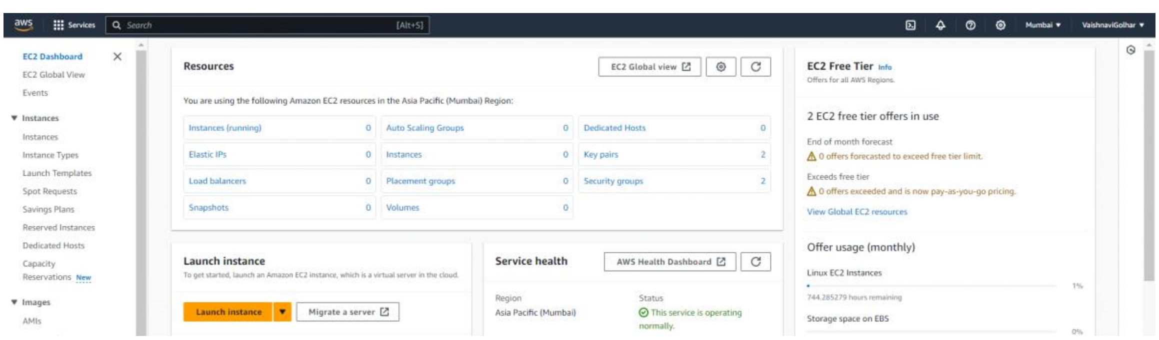
6. Allow traffic from EC2 to RDS instance

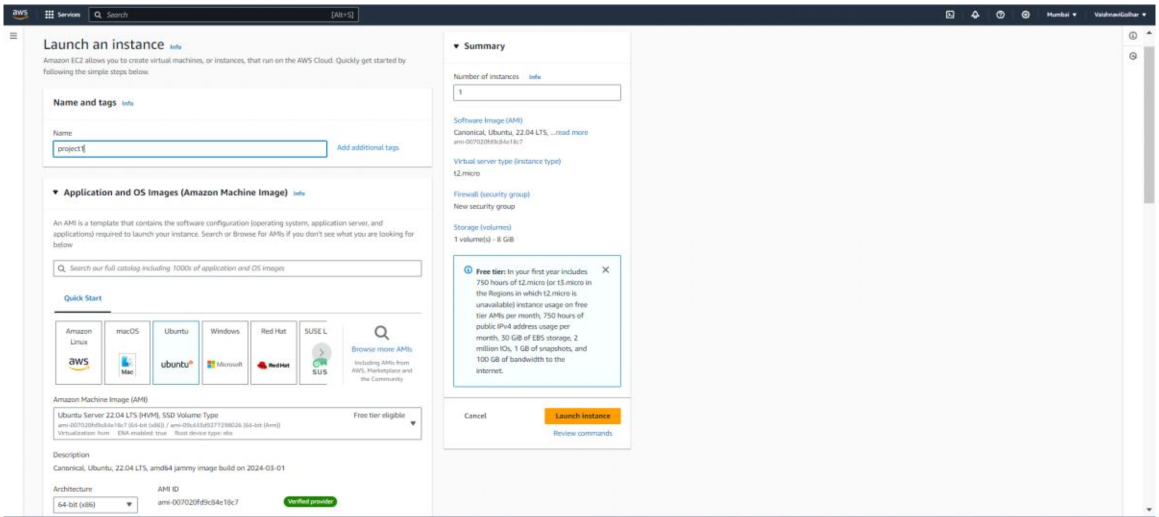
7. Allow all-traffic to EC2 instance

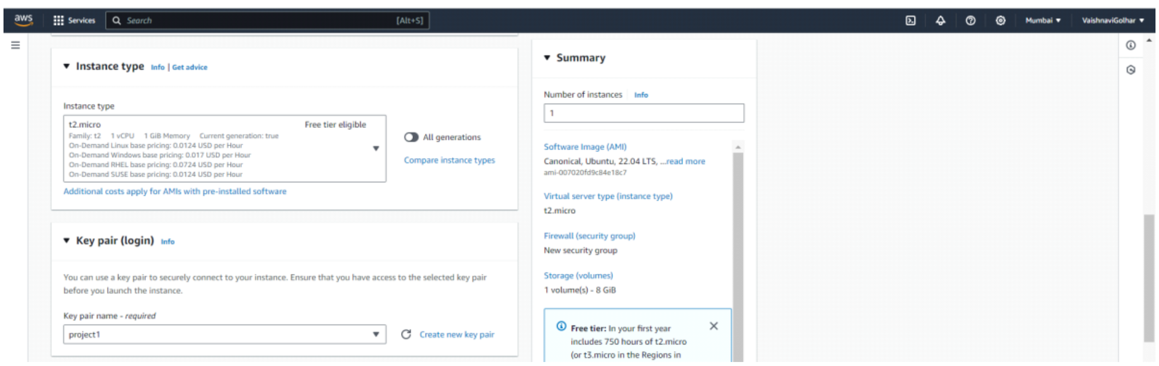
SOLUTION:

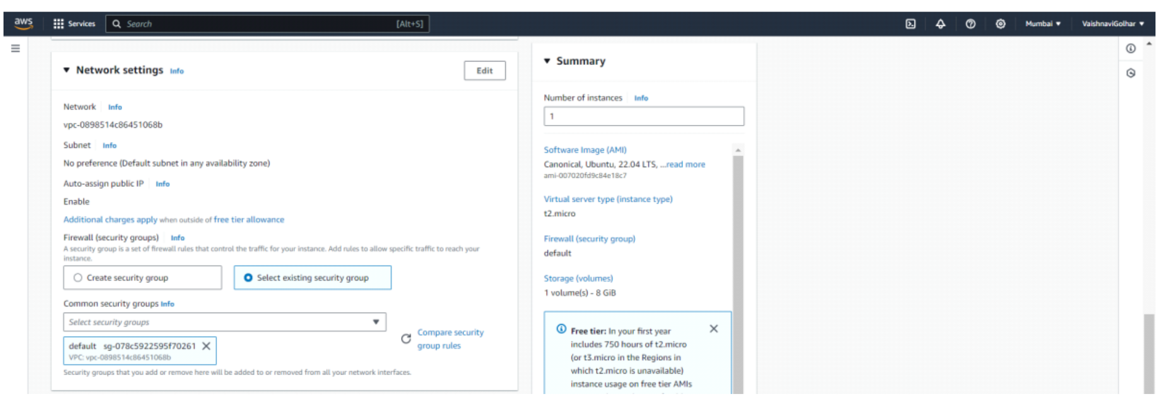
1. Launch EC2 Instances:

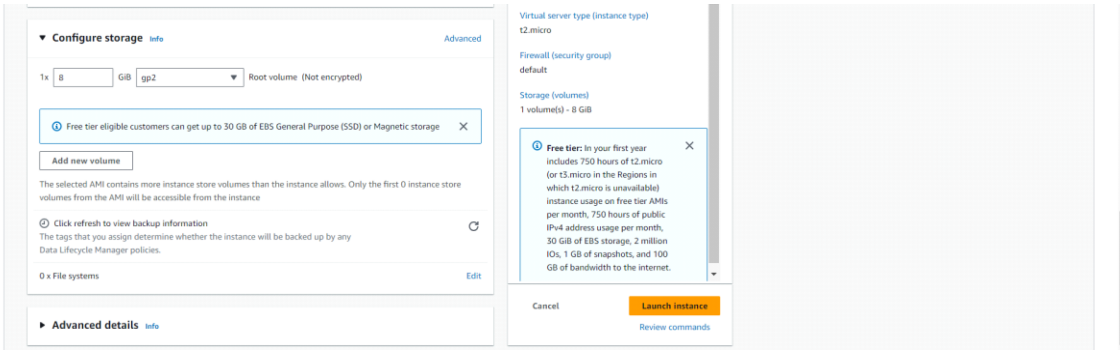
Launch multiple EC2 instances to host your PHP website. Ensure that you choose appropriate instance types based on your workload requirements.

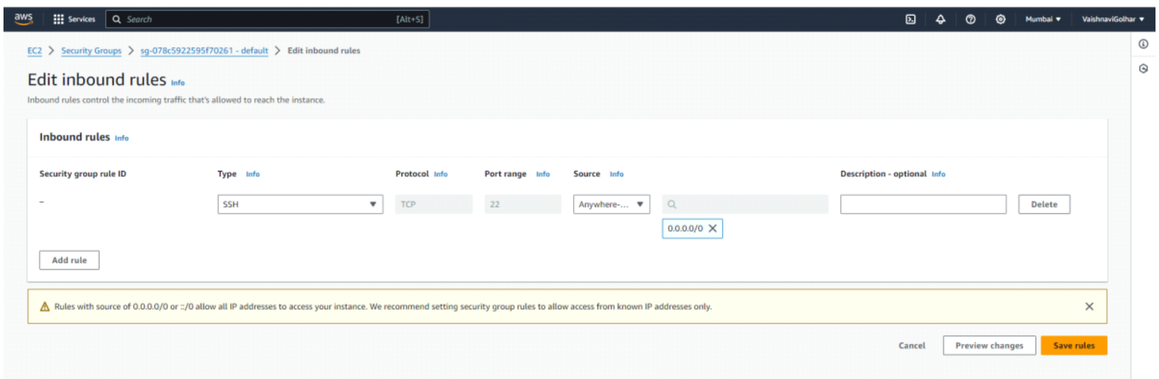


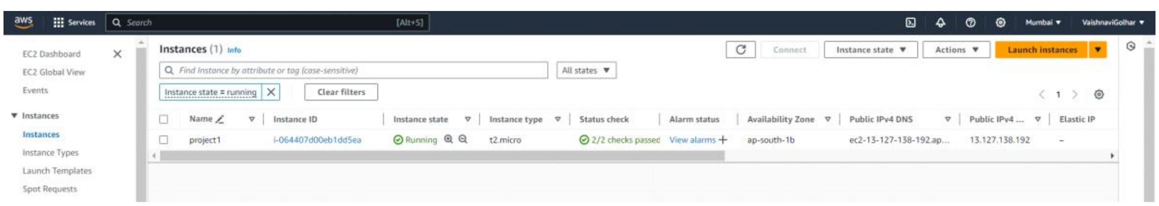

Launch an instance
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags
Name
project1
Add additional tags
Application and OS Images (Amazon Machine Image)
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
Browse more AMIs
Amazon Machine Image (AMI)
Ubuntu Server 22.04 LTS (HVM), SSD Volume Type
Free tier eligible
Description
Canonical, Ubuntu, 22.04 LTS, amd64 jammy image build on 2024-03-01
Architecture
64-bit (x86)
AMI ID
ami-007020fd9c84e18c7
Verified provider
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Review commands


Instance type
Info | Get advice
Instance type
t2.micro
Free tier eligible
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0724 USD per Hour
On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login)
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
project1
Create new key pair
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
ami-007020fd9c84e18c7
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in


Network settings
Edit
Network
vpc-0898514c86451068b
Subnet
No preference (Default subnet in any availability zone)
Auto-assign public IP
Enable
Additional charges apply when outside of free tier allowance
Firewall (security groups)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups
Select security groups
default sg-078c5922595f70261
VPC: vpc-0898514c86451068b
Compare security group rules
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
ami-007020fd9c84e18c7
Virtual server type (instance type)
t2.micro
Firewall (security group)
default
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs







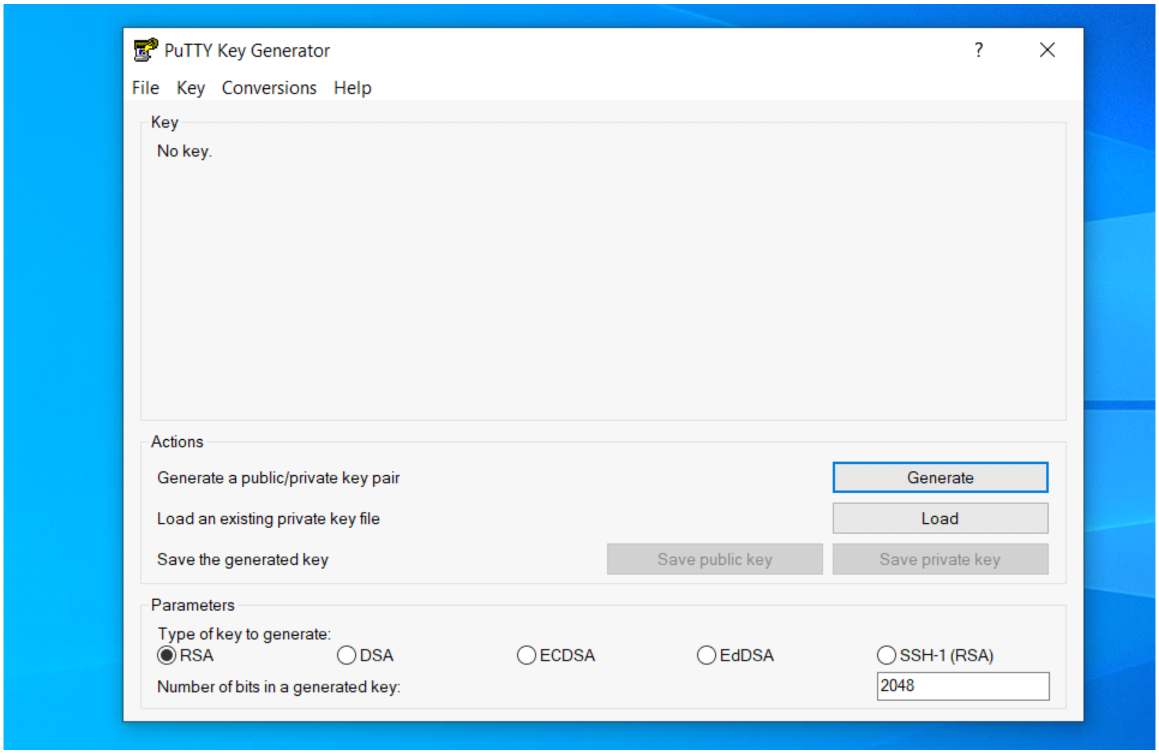
2. Connect your system with your EC2 instance:

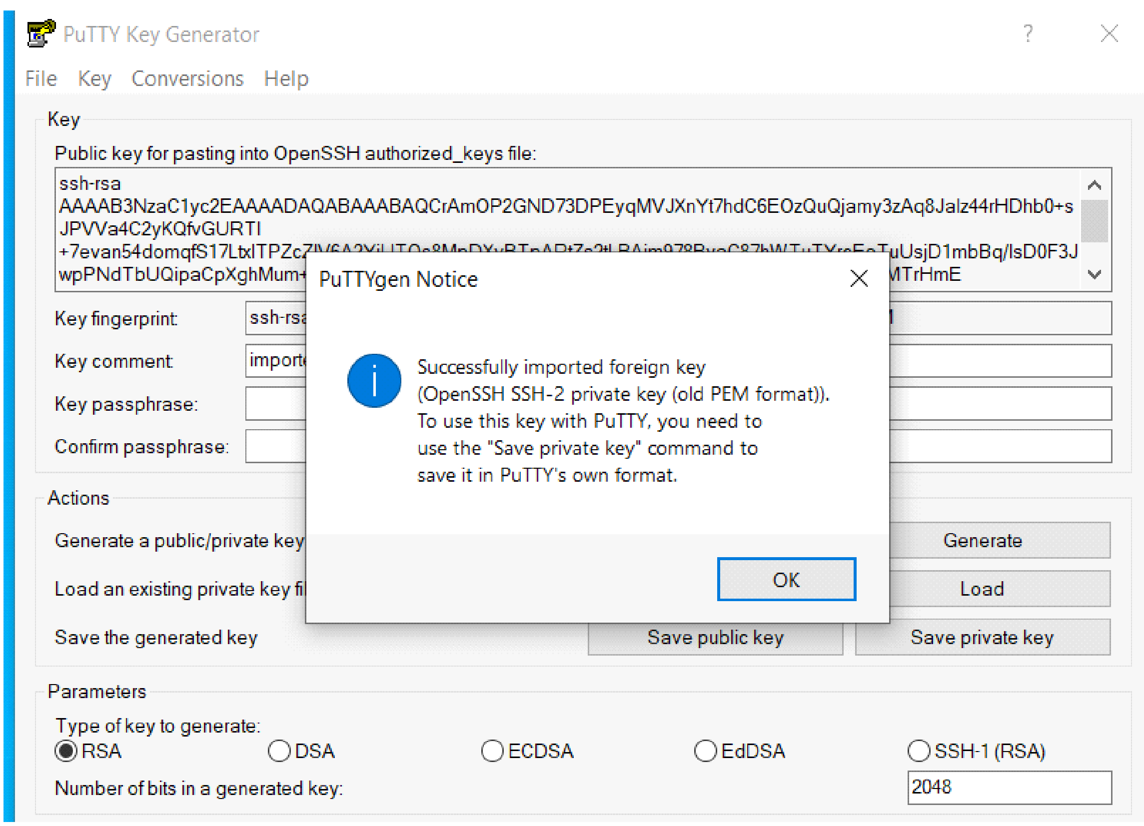
a. First you need to install PuTTY on your system and then

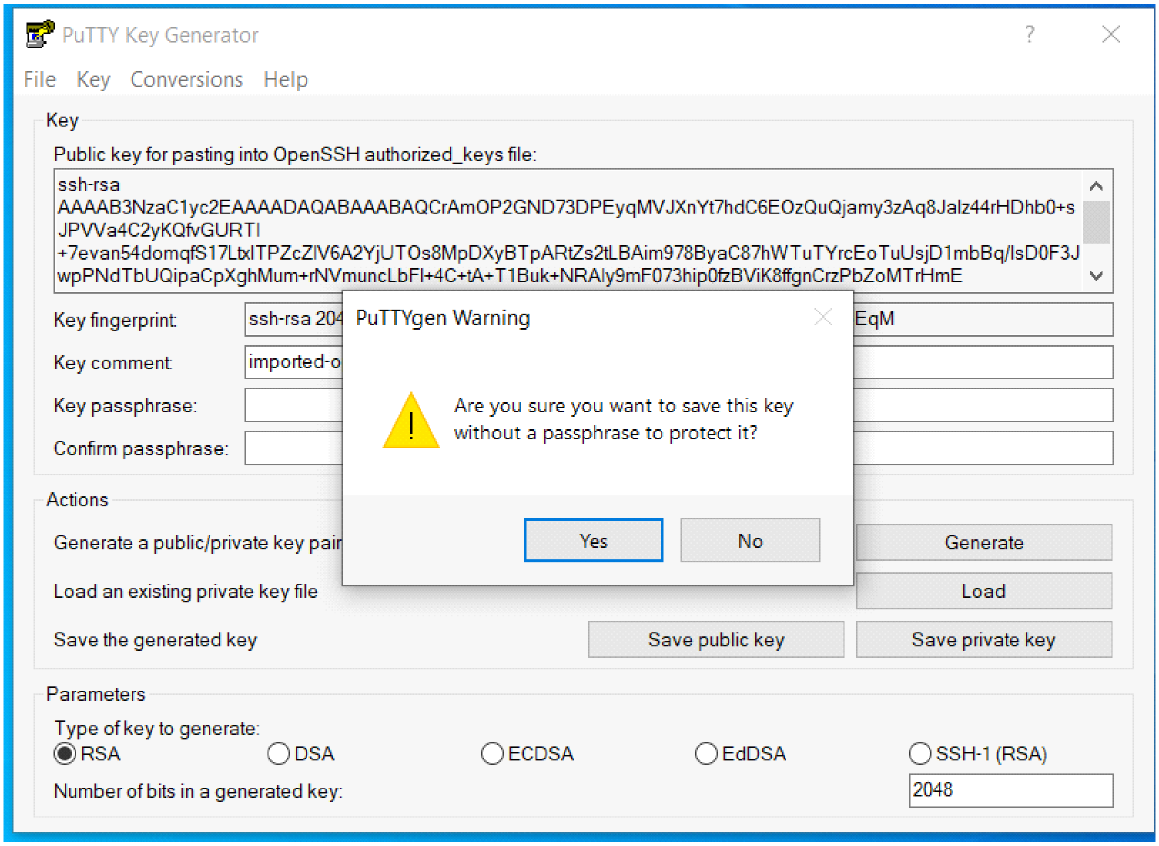
Now it’s time to convert your private key using PuTTYgen

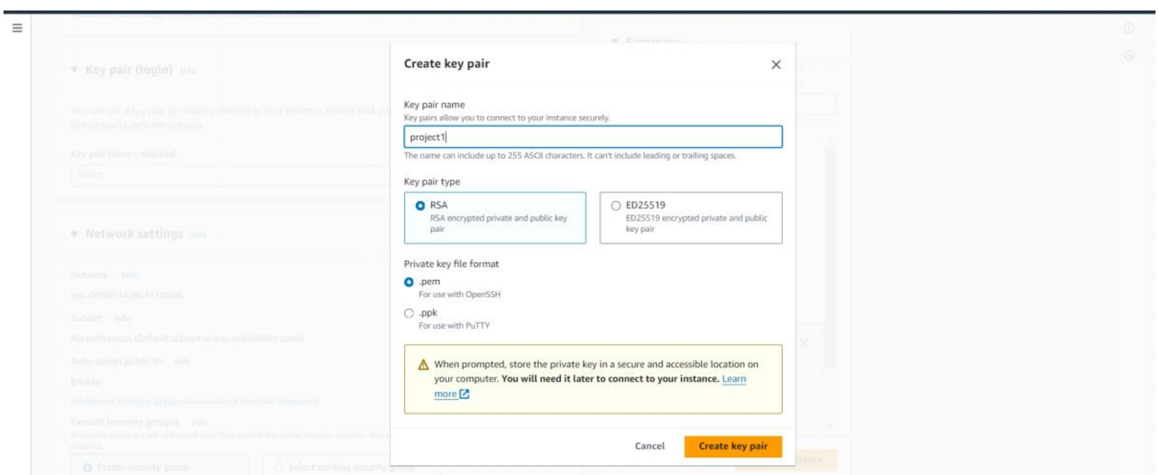
PuTTY won’t be able to support this .pem file, so you’d require a PuTTY gen

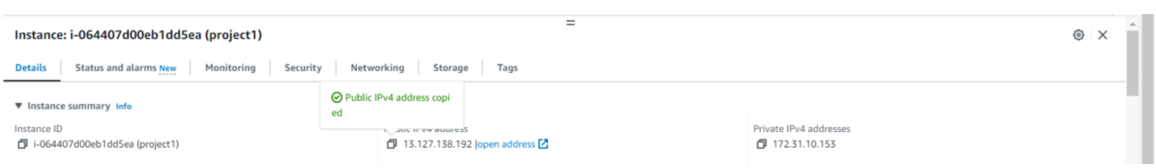
tool which can convert your .pem file into .ppk format, because you need a .ppk file in order to connect it with your instance

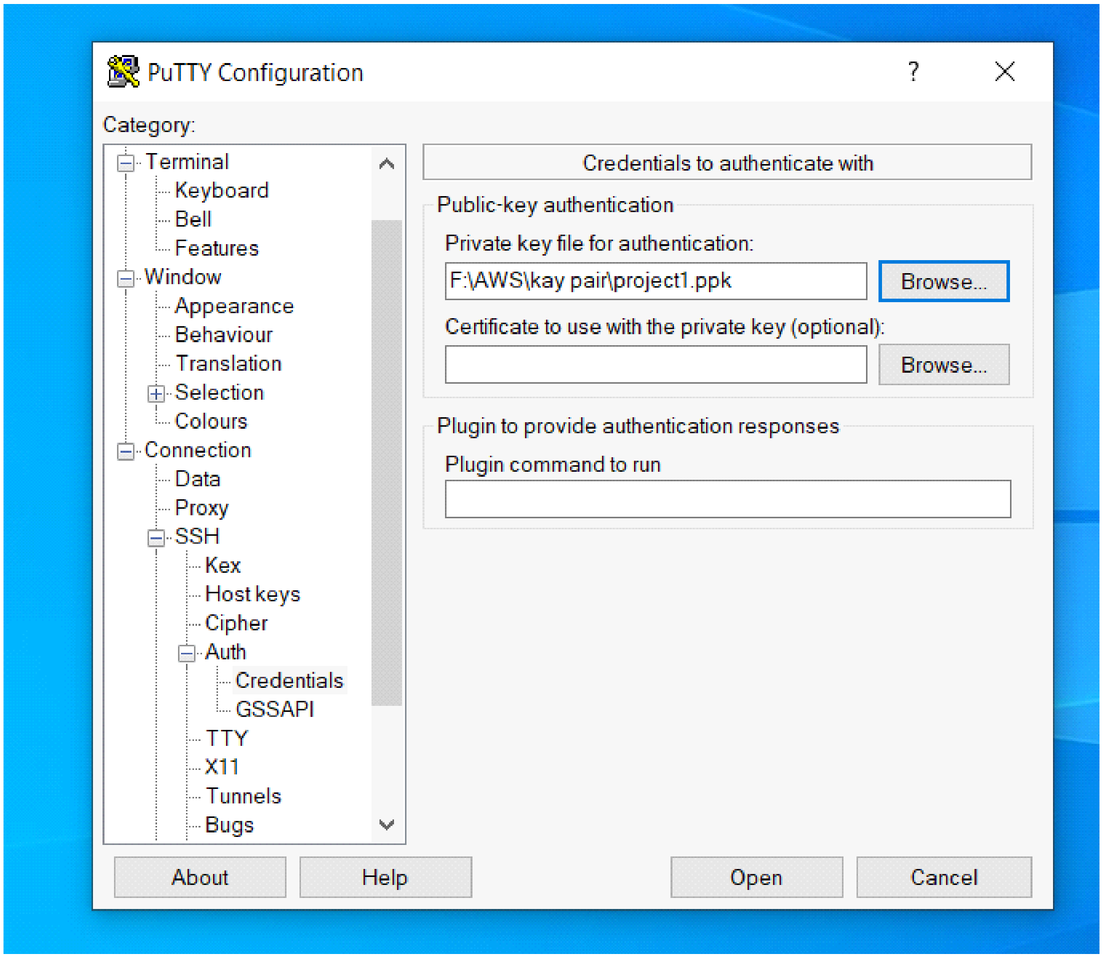

PuTTY Key Generator
File Key Conversions Help
Key
No key.
Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048

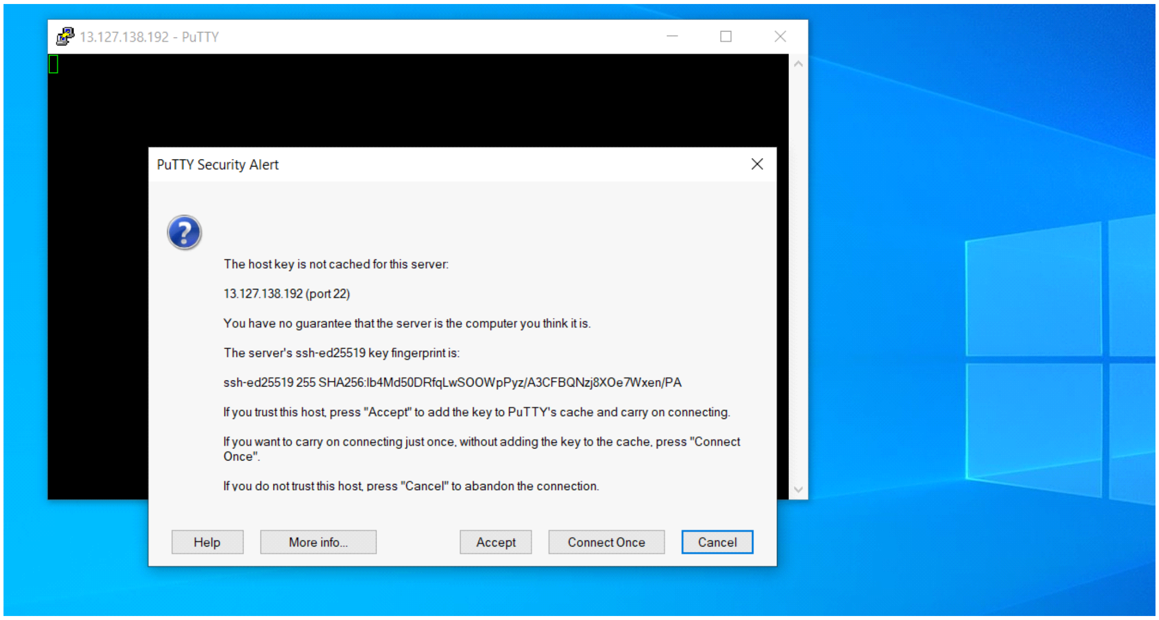

PuTTY Key Generator
File Key Conversions Help
Key
Public key for pasting into OpenSSH authorized_keys file:
Key fingerprint
Key comment
Key passphrase:
Confirm passphrase:
PuTTYgen Notice
Successfully imported foreign key
(OpenSSH SSH-2 private key (old PEM format)).
To use this key with PuTTY, you need to
use the "Save private key" command to
save it in PuTTY's own format.
OK
Actions
Generate
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048

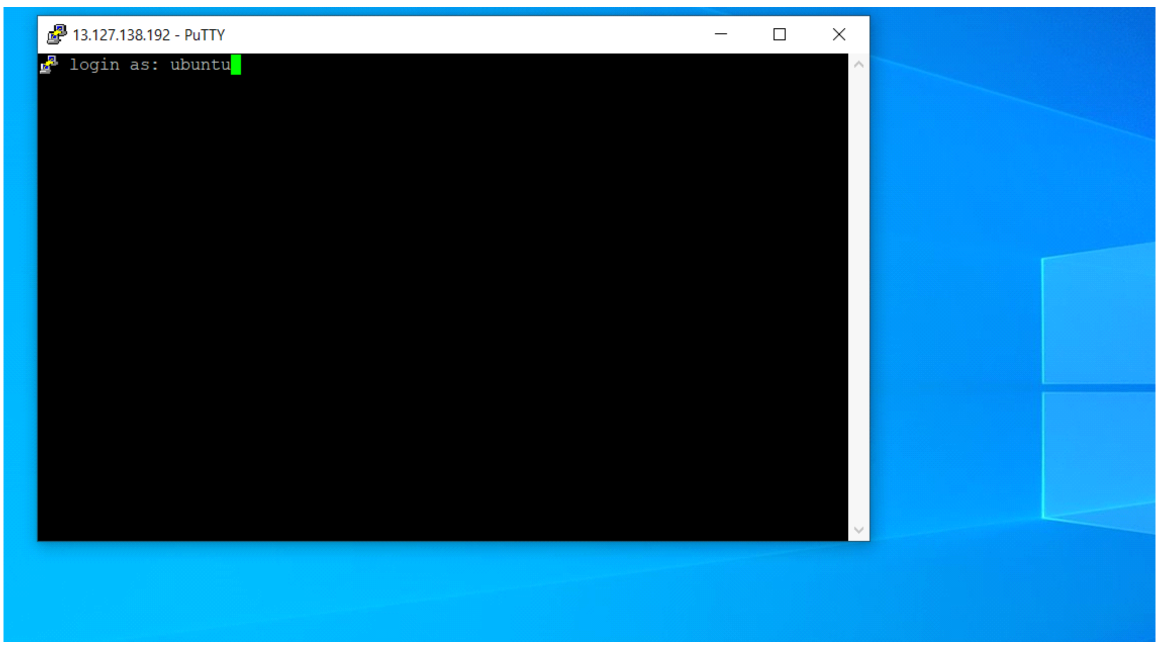

PuTTY Key Generator
File
Key
Conversions
Help
Key
Public key for pasting into OpenSSH authorized_keys file:
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQCrAmOP2GND73DPEyqMVJXnYt7hdC6EOzQuQjamy3zAq8Jalz44rHDhb0+s
JPVVa4C2yKQfvGURTI
+7evan54domqfS17LtxITPZcZlV6A2YjUTOs8MpDXyBTpARtZs2tLBAim978ByaC87hWTuTYrcEoTuUsjD1mbBq/lsD0F3J
wpPNdTbUQipaCpXghMum+rNVmuncLbFl+4C+tA+T1Buk+NRAly9mF073hip0fzBViK8ffgnCrzPbZoMTrHmE
Key fingerprint
ssh-rsa 204
EqM
Key comment
imported-o
Key passphrase:
Confirm passphrase:
PuTTYgen Warning
Are you sure you want to save this key without a passphrase to protect it?
Yes
No
Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048

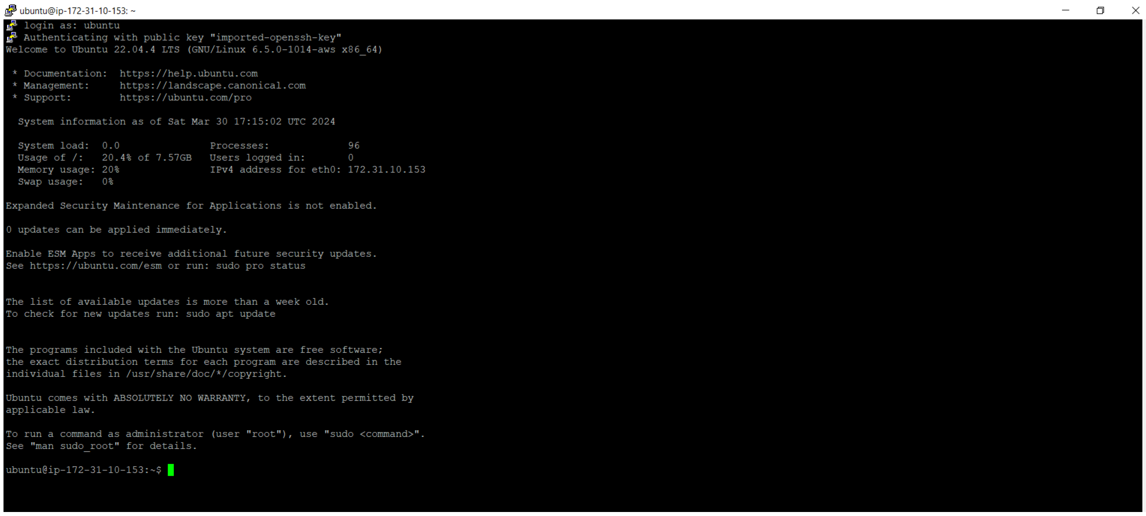

Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
project1
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair


Instance: i-064407d00eb1dd5ea (project1)
Details
Status and alarms New
Monitoring
Security
Networking
Storage
Tags
Public IPv4 address copied
Instance summary Info
Instance ID
i-064407d00eb1dd5ea (project1)
13.127.138.192 |open address
Private IPv4 addresses
172.31.10.153


PuTTY Configuration
Category:
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Bugs
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
F:\AWS\kay pair\project1.ppk
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel


13.127.138.192 - PuTTY
PuTTY Security Alert
The host key is not cached for this server:
13.127.138.192 (port 22)
You have no guarantee that the server is the computer you think it is.
The server's ssh-ed25519 key fingerprint is:
ssh-ed25519 255 SHA256:lb4Md50DRfqLwSOOWpPyz/A3CFBQNzj8XOe7Wxen/PA
If you trust this host, press "Accept" to add the key to PuTTY's cache and carry on connecting.
If you want to carry on connecting just once, without adding the key to the cache, press "Connect Once".
If you do not trust this host, press "Cancel" to abandon the connection.
Help
More info...
Accept
Connect Once
Cancel


13.127.138.192 - PuTTY
login as: ubuntu



b. First Update your system using the command

1. sudo apt-get update

2. Then use this command in PuTTY to install Apache2

    sudo apt-get install apache2

3. Then install php-mysql using the following command

    sudo add-apt-repository -y ppa:ondrej/php

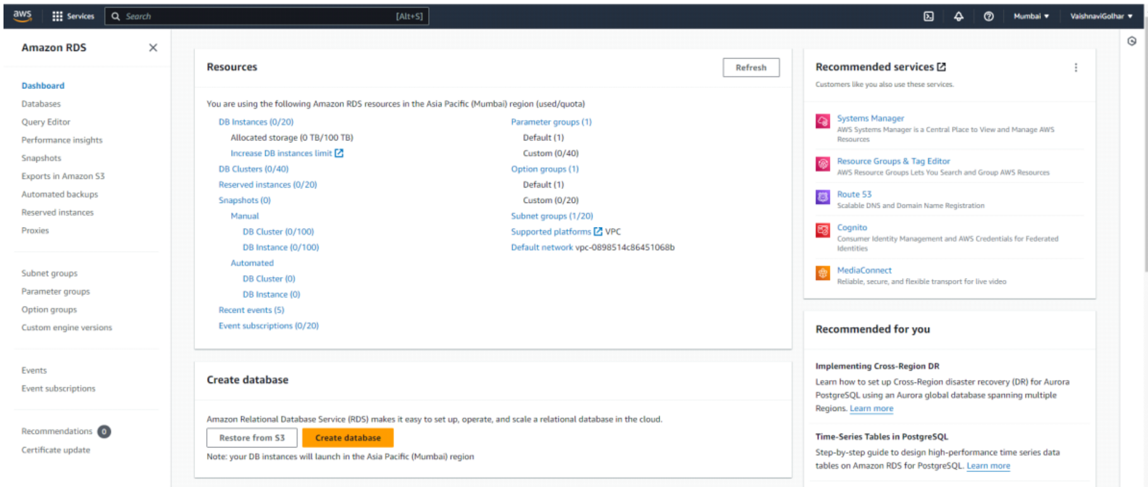
    sudo apt install php5.6 mysql-client php5.6-mysqli

Now everything is updated in your system

3. Create RDS Instance:

Provision an Amazon RDS (Relational Database Service) instance to host your MySQL database.

Choose appropriate instance specifications based on your database workload and storage requirements.


Services
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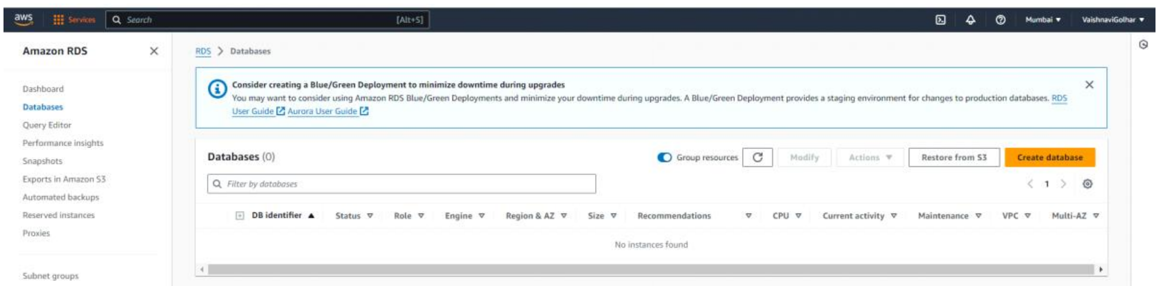
Amazon RDS
Dashboard
Databases
Query Editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Events
Event subscriptions
Recommendations
Certificate update
Resources
Refresh
You are using the following Amazon RDS resources in the Asia Pacific (Mumbai) region (used/quota)
DB Instances (0/20)
Allocated storage (0 TB/100 TB)
Increase DB instances limit
DB Clusters (0/40)
Reserved instances (0/20)
Snapshots (0)
Manual
DB Cluster (0/100)
DB Instance (0/100)
Automated
DB Cluster (0)
DB Instance (0)
Recent events (5)
Event subscriptions (0/20)
Parameter groups (1)
Default (1)
Custom (0/40)
Option groups (1)
Default (1)
Custom (0/20)
Subnet groups (1/20)
Supported platforms VPC
Default network vpc-0898514c86451068b
Create database
Amazon Relational Database Service (RDS) makes it easy to set up, operate, and scale a relational database in the cloud.
Restore from S3
Create database
Note: your DB instances will launch in the Asia Pacific (Mumbai) region
Recommended services
Customers like you also use these services.
Systems Manager
AWS Systems Manager is a Central Place to View and Manage AWS Resources
Resource Groups & Tag Editor
AWS Resource Groups Lets You Search and Group AWS Resources
Route 53
Scalable DNS and Domain Name Registration
Cognito
Consumer Identity Management and AWS Credentials for Federated Identities
MediaConnect
Reliable, secure, and flexible transport for live video
Recommended for you
Implementing Cross-Region DR
Learn how to set up Cross-Region disaster recovery (DR) for Aurora PostgreSQL using an Aurora global database spanning multiple Regions. Learn more
Time-Series Tables in PostgreSQL
Step-by-step guide to design high-performance time series data tables on Amazon RDS for PostgreSQL. Learn more


Services
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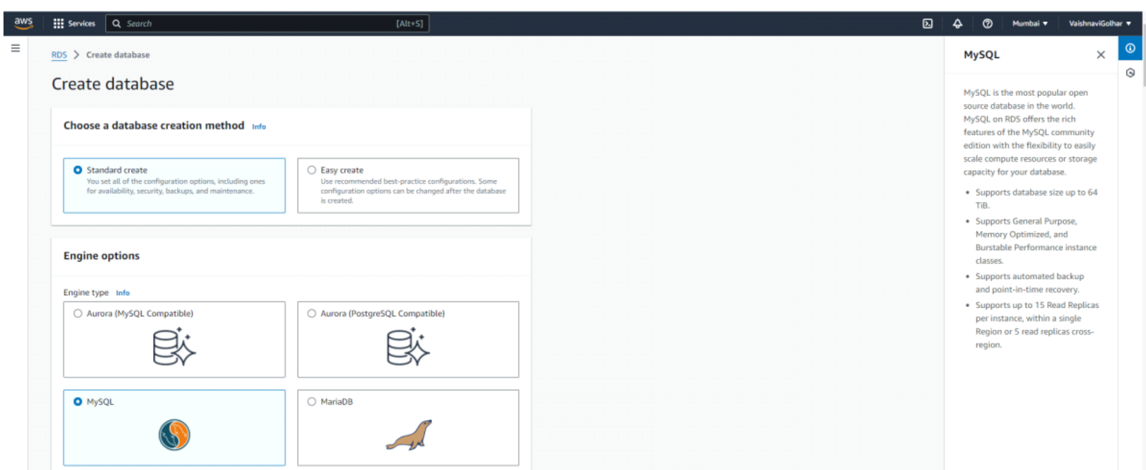
Amazon RDS
Dashboard
Databases
Query Editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
RDS > Databases
Consider creating a Blue/Green Deployment to minimize downtime during upgrades
You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. RDS User Guide Aurora User Guide
Databases (0)
Group resources
Modify
Actions
Restore from S3
Create database
Filter by databases
DB identifier
Status
Role
Engine
Region & AZ
Size
Recommendations
CPU
Current activity
Maintenance
VPC
Multi-AZ
No instances found


Services
Search
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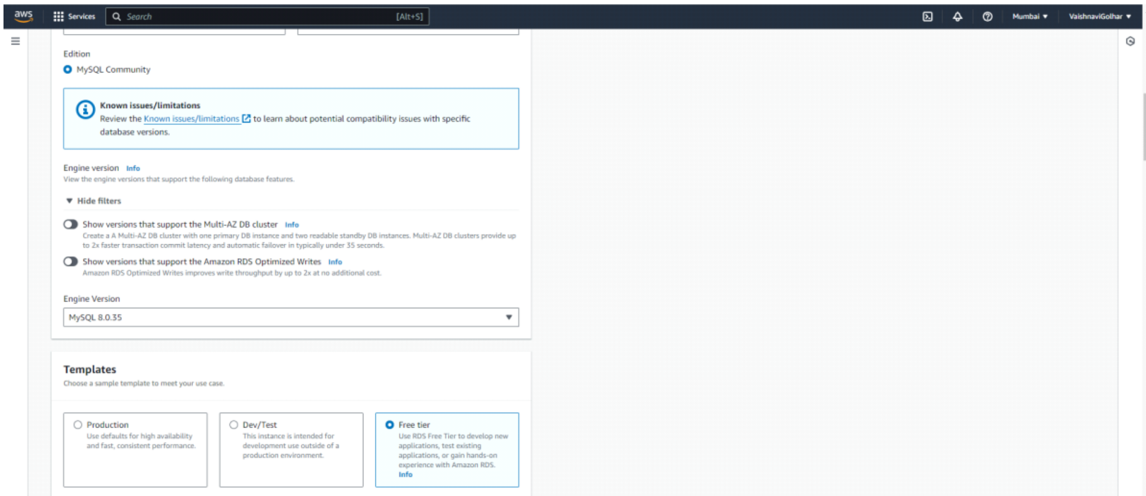
RDS > Create database
Create database
Choose a database creation method
Info
Standard create
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.
Engine options
Engine type
Info
Aurora (MySQL Compatible)
Aurora (PostgreSQL Compatible)
MySQL
MariaDB
MySQL
MySQL is the most popular open source database in the world. MySQL on RDS offers the rich features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.
Supports database size up to 64 TiB.
Supports General Purpose, Memory Optimized, and Burstable Performance instance classes.
Supports automated backup and point-in-time recovery.
Supports up to 15 Read Replicas per instance, within a single Region or 5 read replicas cross-region.


Services
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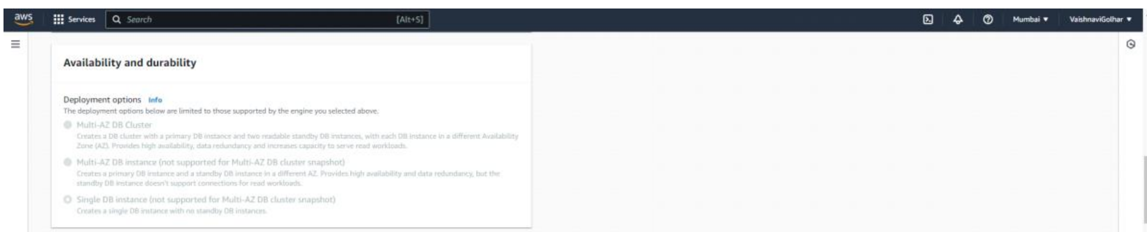
Edition
MySQL Community
Known issues/limitations
Review the Known issues/limitations to learn about potential compatibility issues with specific database versions.
Engine version Info
View the engine versions that support the following database features.
Hide filters
Show versions that support the Multi-AZ DB cluster Info
Create a A Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.
Show versions that support the Amazon RDS Optimized Writes Info
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
Engine Version
MySQL 8.0.35
Templates
Choose a sample template to meet your use case.
Production
Use defaults for high availability and fast, consistent performance.
Dev/Test
This instance is intended for development use outside of a production environment.
Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.
Info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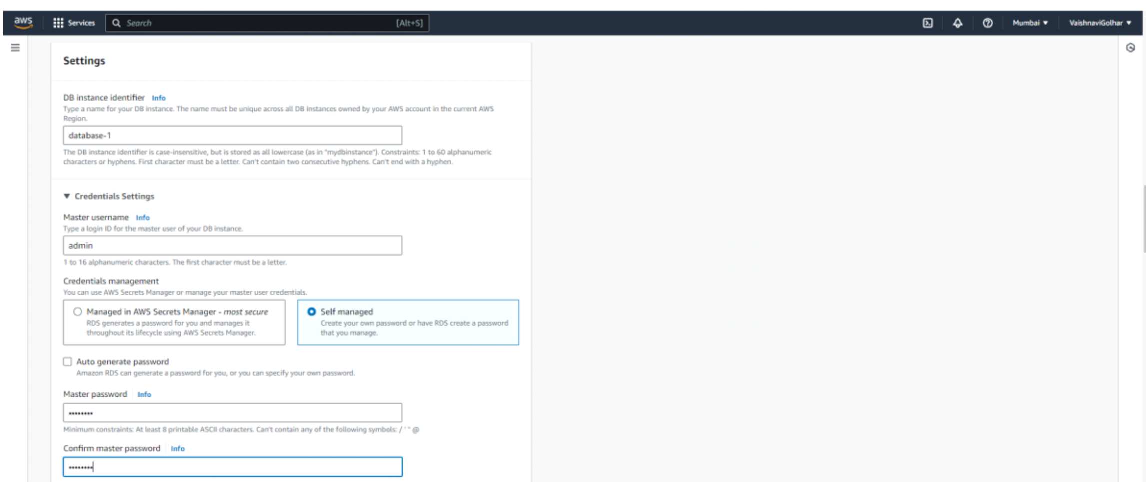
Availability and durability
Deployment options Info
The deployment options below are limited to those supported by the engine you selected above.
Multi-AZ DB Cluster
Creates a DB cluster with a primary DB instance and two readable standby DB instances, with each DB instance in a different Availability Zone (AZ). Provides high availability, data redundancy and increases capacity to serve read workloads.
Multi-AZ DB instance (not supported for Multi-AZ DB cluster snapshot)
Creates a primary DB instance and a standby DB instance in a different AZ. Provides high availability and data redundancy, but the standby DB instance doesn't support connections for read workloads.
Single DB instance (not supported for Multi-AZ DB cluster snapshot)
Creates a single DB instance with no standby DB instances.


Services
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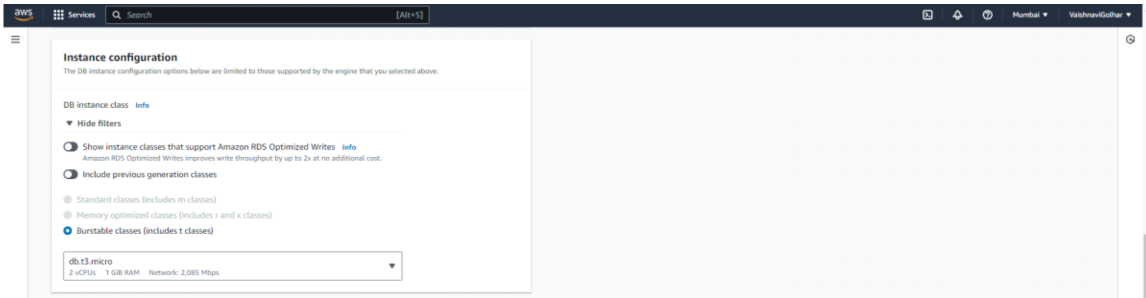
Settings
DB instance identifier Info
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
database-1
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.
Credentials Settings
Master username Info
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.
Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
Self managed
Create your own password or have RDS create a password that you manage.
Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.
Master password Info
Confirm master password Info


aws
Services
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Mumbai
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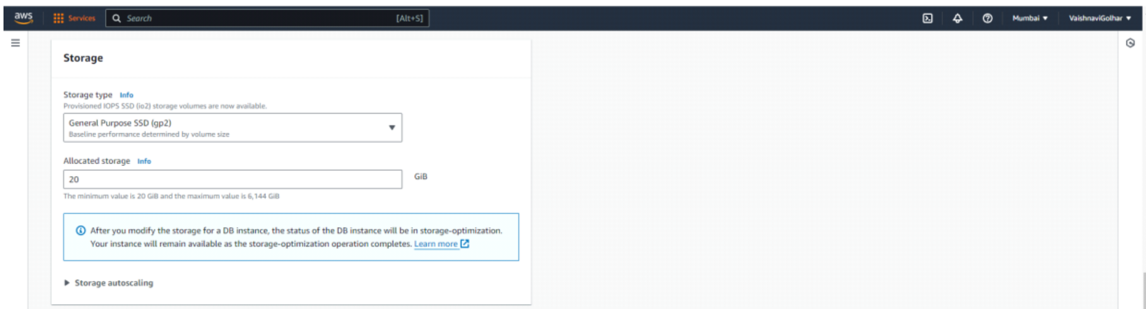
Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.
DB instance class Info
Hide filters
Show instance classes that support Amazon RDS Optimized Writes Info
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
Include previous generation classes
Standard classes (includes m classes)
Memory optimized classes (includes r and x classes)
Burstable classes (includes t classes)
db.t3.micro
2 vCPUs 1 GiB RAM Network: 2,085 Mbps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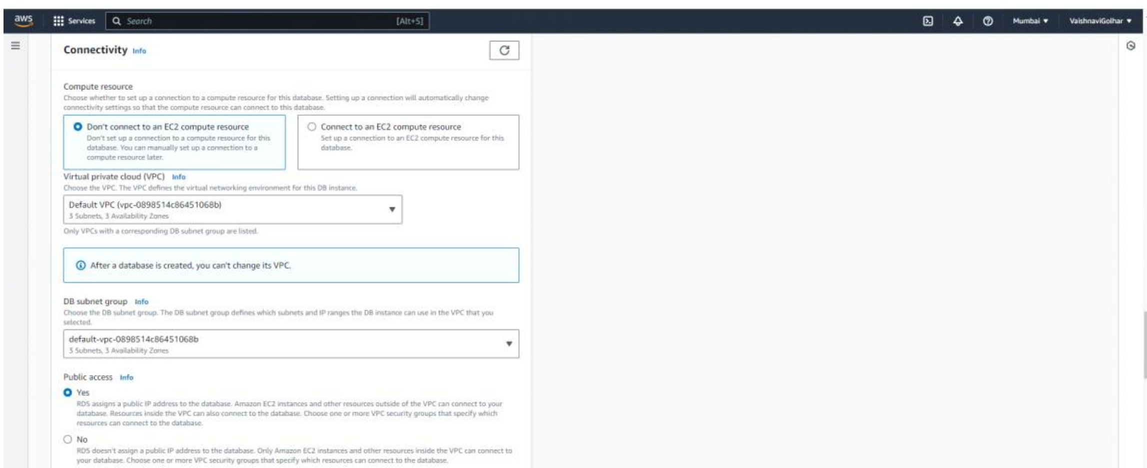
Storage
Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size
Allocated storage Info
20
GiB
The minimum value is 20 GiB and the maximum value is 6,144 GiB
After you modify the storage for a DB instance, the status of the DB instance will be in storage-optimization. Your instance will remain available as the storage-optimization operation completes. Learn more
Storage autoscaling


aws
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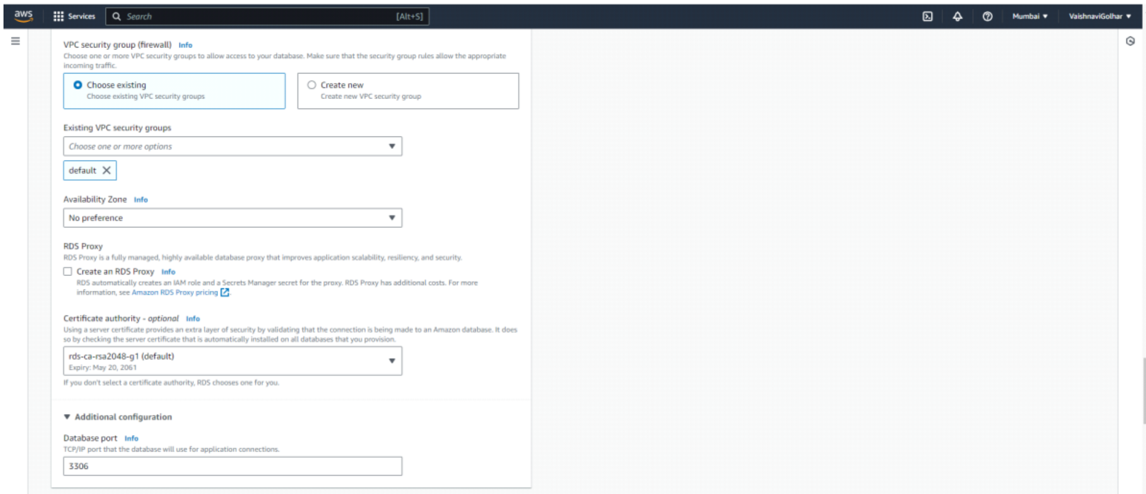
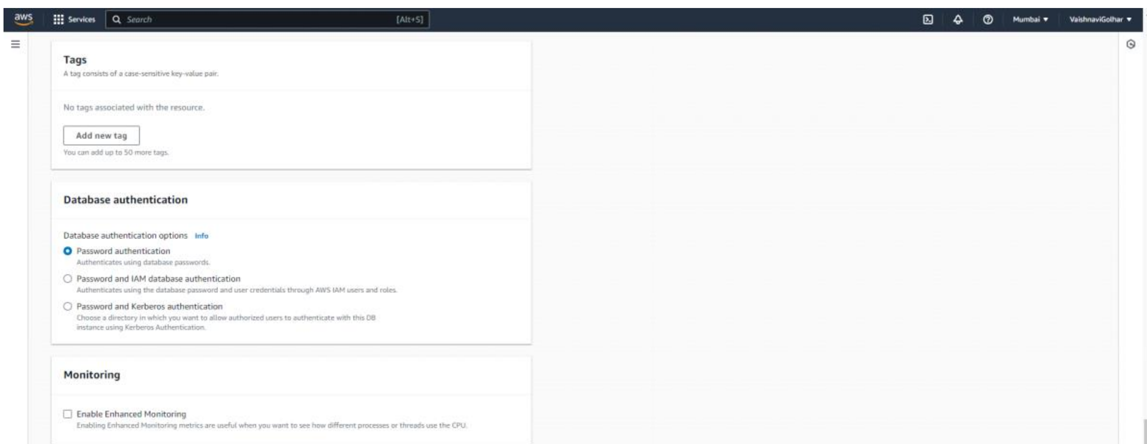
Connectivity Info
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
Default VPC (vpc-0898514c86451068b)
3 Subnets, 3 Availability Zones
Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
default-vpc-0898514c86451068b
3 Subnets, 3 Availability Zones
Public access Info
Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.


aws
Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
Existing VPC security groups
Choose one or more options
default
Availability Zone Info
No preference
RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing
Certificate authority - optional Info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061
If you don't select a certificate authority, RDS chooses one for you.
Additional configuration
Database port Info
TCP/IP port that the database will use for application connections.
3306


aws
Services
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Mumbai
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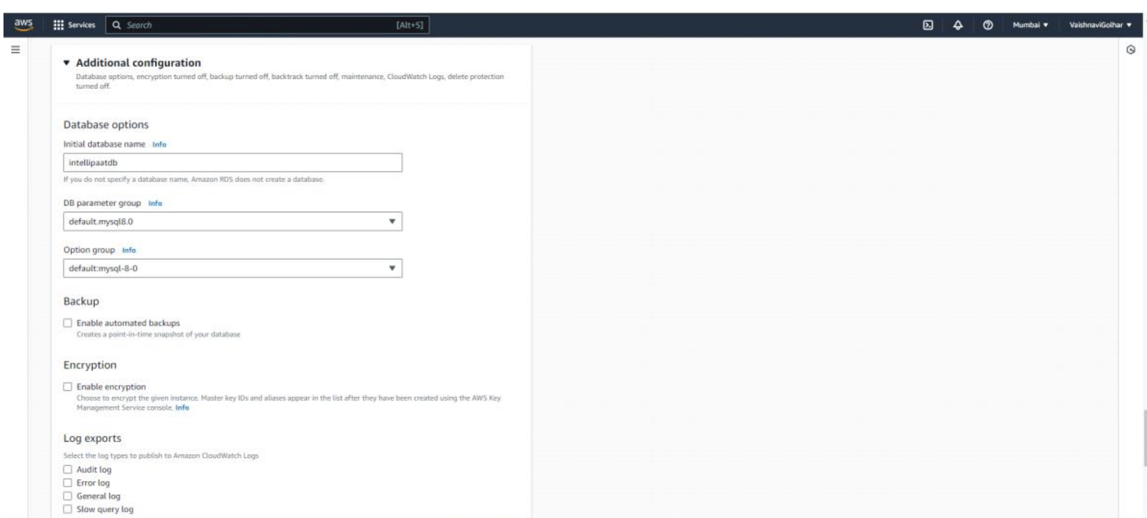
Tags
A tag consists of a case-sensitive key-value pair.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Database authentication
Database authentication options Info
Password authentication
Authenticates using database passwords.
Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.
Monitoring
Enable Enhanced Monitoring
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.


aws
Services
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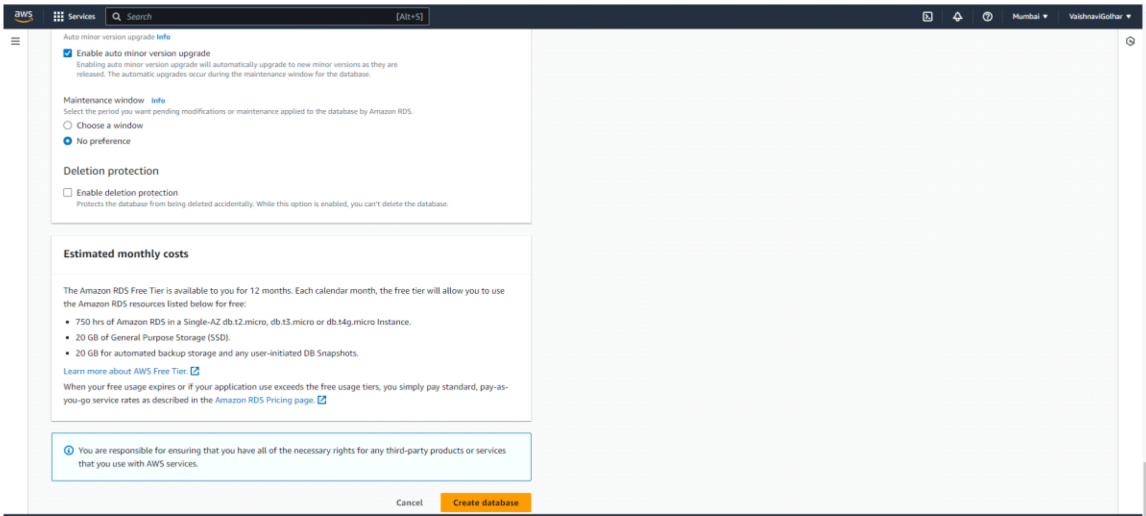
Additional configuration
Database options, encryption turned off, backup turned off, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.
Database options
Initial database name Info
intellipaatdb
If you do not specify a database name, Amazon RDS does not create a database.
DB parameter group Info
default.mysql8.0
Option group Info
default:mysql-8-0
Backup
Enable automated backups
Creates a point-in-time snapshot of your database
Encryption
Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. Info
Log exports
Select the log types to publish to Amazon CloudWatch Logs
Audit log
Error log
General log
Slow query log


Services
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Mumbai
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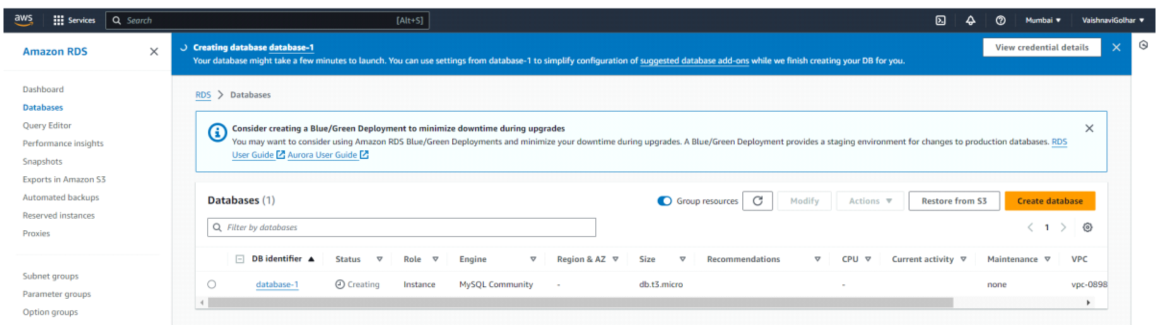
Auto minor version upgrade Info
Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.
Maintenance window Info
Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.
Choose a window
No preference
Deletion protection
Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.
Estimated monthly costs
The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:
750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
20 GB of General Purpose Storage (SSD).
20 GB for automated backup storage and any user-initiated DB Snapshots.
Learn more about AWS Free Tier.
When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the Amazon RDS Pricing page.
You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.
Cancel
Create database


Services
Search
[Alt+S]
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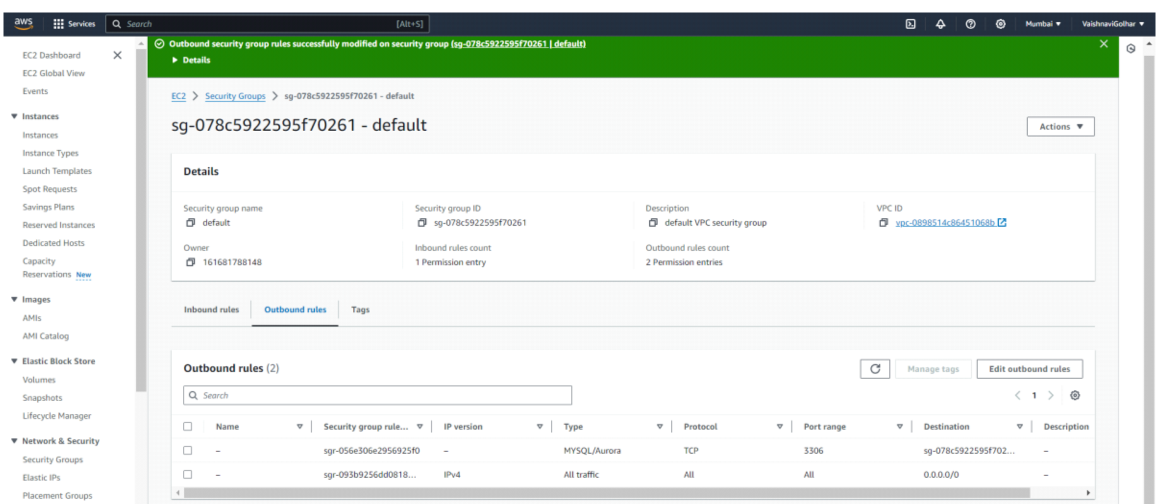
Amazon RDS
Creating database database-1
Your database might take a few minutes to launch. You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.
View credential details
Dashboard
Databases
Query Editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
RDS > Databases
Consider creating a Blue/Green Deployment to minimize downtime during upgrades
You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. RDS User Guide Aurora User Guide
Databases (1)
Group resources
Modify
Actions
Restore from S3
Create database
Filter by databases
DB identifier Status Role Engine Region & AZ Size Recommendations CPU Current activity Maintenance VPC
database-1 Creating Instance MySQL Community - db.t3.micro - none vpc-0898


Services
Search
[Alt+S]
Mumbai
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Outbound security group rules successfully modified on security group (sg-078c5922595f70261 | default)
Details
EC2 Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
EC2 > Security Groups > sg-078c5922595f70261 - default
sg-078c5922595f70261 - default
Actions
Details
Security group name
default
Security group ID
sg-078c5922595f70261
Description
default VPC security group
VPC ID
vpc-0898514c86451068b
Owner
161681788148
Inbound rules count
1 Permission entry
Outbound rules count
2 Permission entries
Inbound rules
Outbound rules
Tags
Outbound rules (2)
Manage tags
Edit outbound rules
Search
Name Security group rule... IP version Type Protocol Port range Destination Description
- sgr-056e306e2956925f0 - MYSQL/Aurora TCP 3306 sg-078c5922595f702... -
- sgr-093b9256dd0818... IPv4 All traffic All All 0.0.0.0/0 -

```
ubuntu@ip-172-31-10-153:~$ mysql -h database-1.cfqs0oommlil.ap-south-1.rds.amazo
naws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 27
Server version: 8.0.35 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```
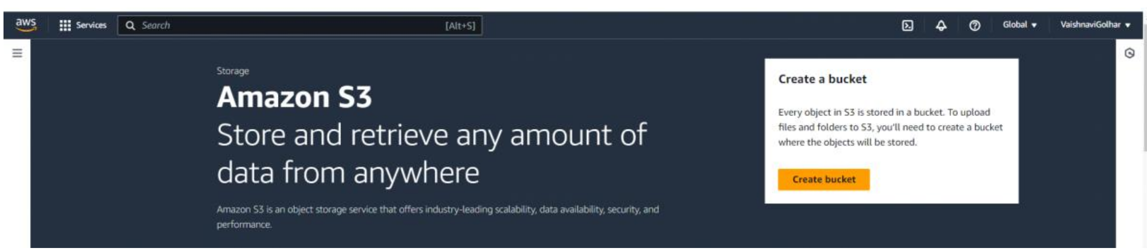
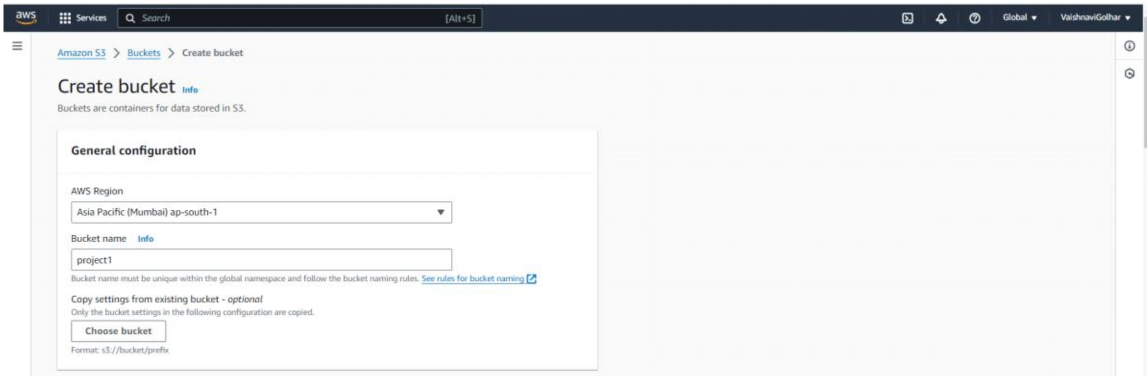

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| intellipaatdb      |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)

mysql>
```

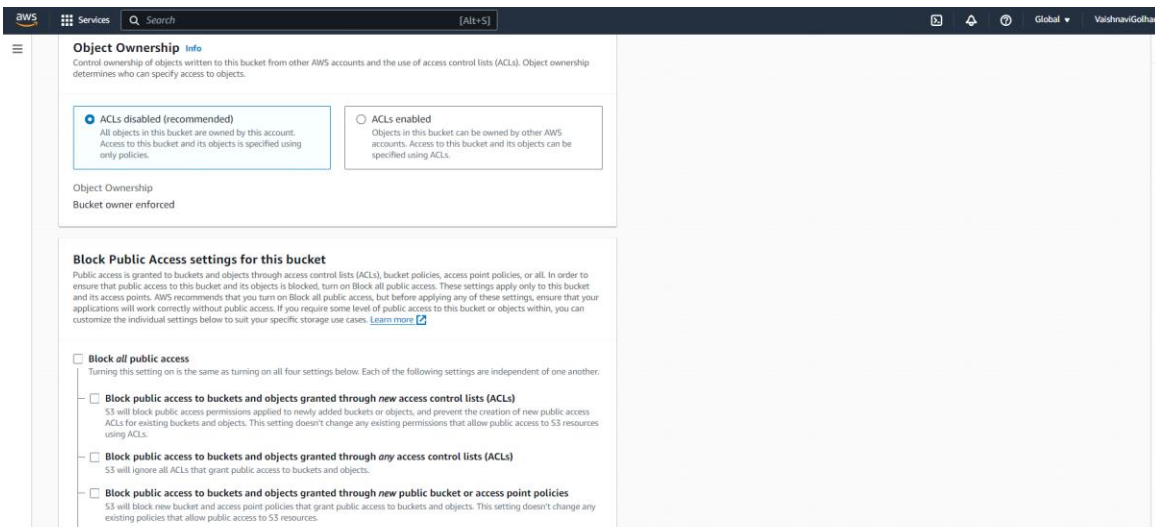
4. Upload the index.php in S3 bucket

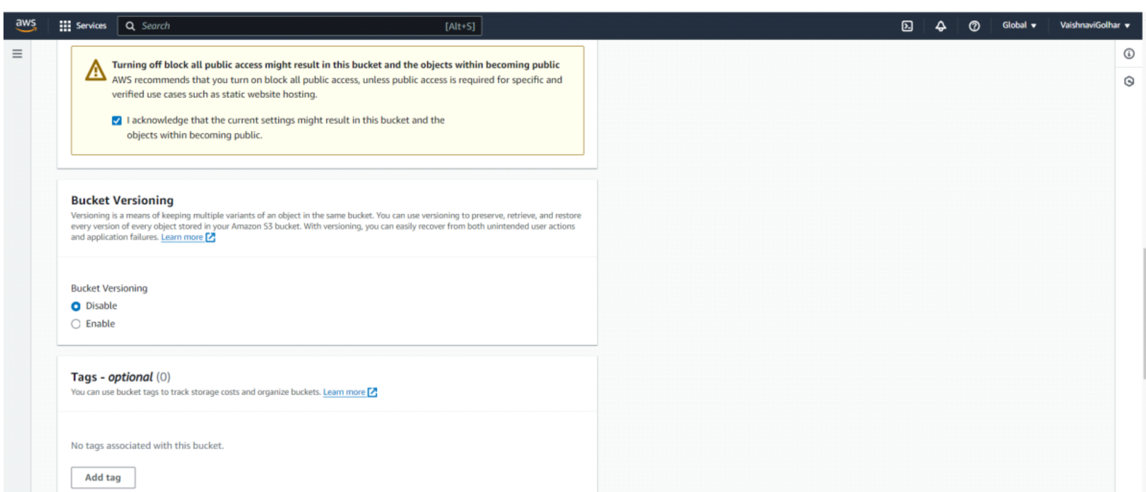
a. Remove the index.html file

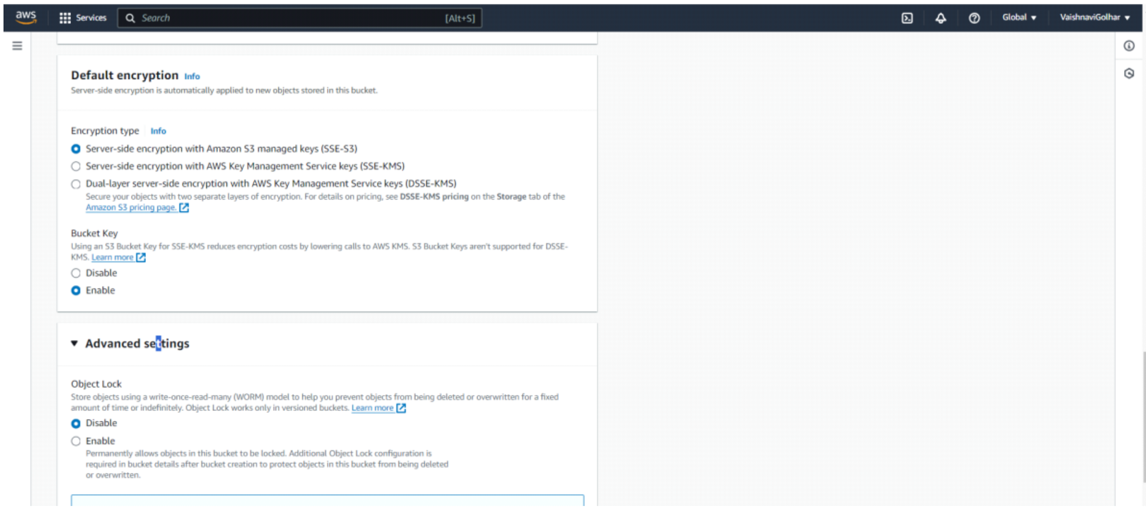
```
ubuntu@ip-172-31-10-153:~$ cd /var/www/html
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.html
ubuntu@ip-172-31-10-153:/var/www/html$ sudo rm index.html
ubuntu@ip-172-31-10-153:/var/www/html$ ls
ubuntu@ip-172-31-10-153:/var/www/html$ sudo touch index.php
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php
ubuntu@ip-172-31-10-153:/var/www/html$
```

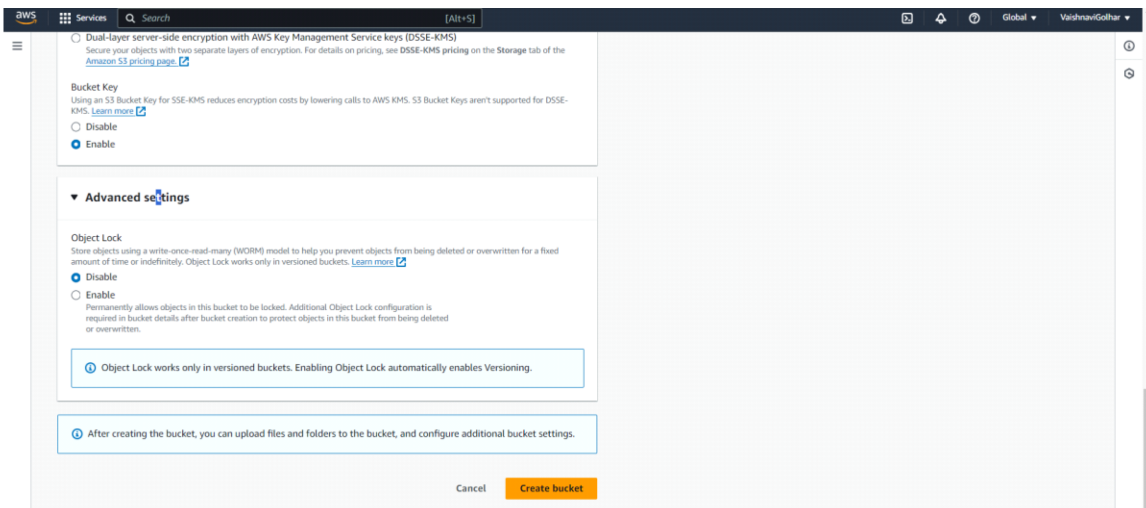


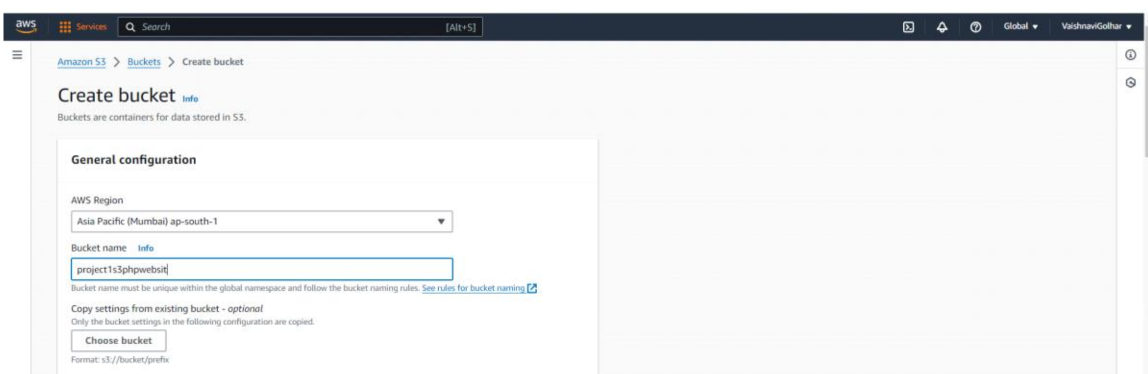

Services
Search
[Alt+S]
Global
VaishnaviGolhar
Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
Asia Pacific (Mumbai) ap-south-1
Bucket name Info
project1
Bucket name must be unique within the global namespace and follow the bucket naming rules. See rules for bucket naming
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix

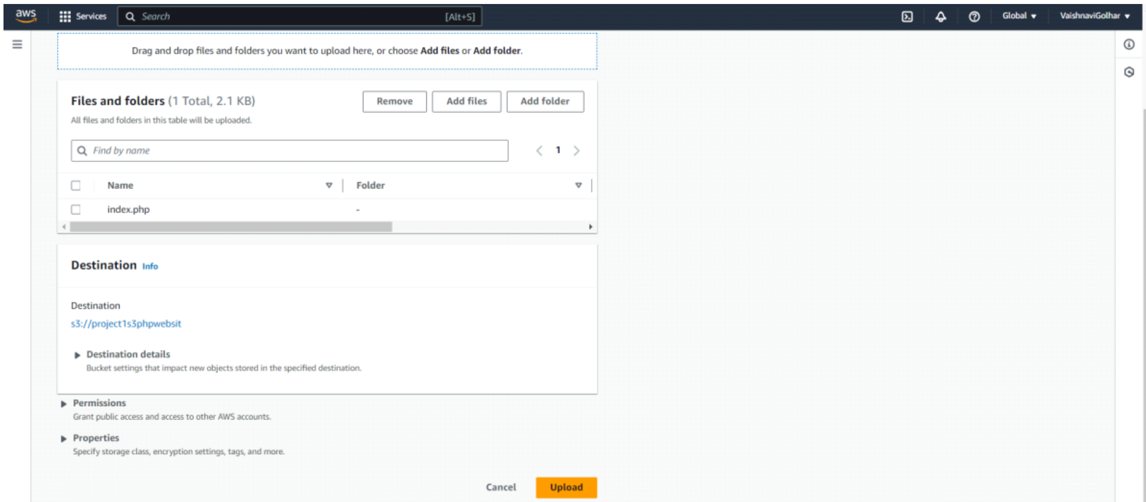

Services
Search
[Alt+S]
Global
Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.
Object Ownership
Bucket owner enforced
Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will ignore all ACLs that grant public access to buckets and objects.
Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

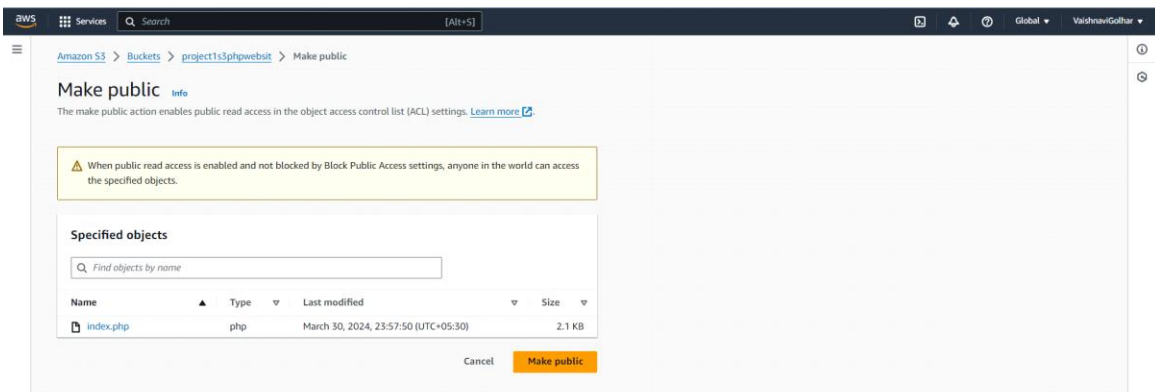

Services
Search
[Alt+S]
Global
VaishnaviGolhar
Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.
I acknowledge that the current settings might result in this bucket and the objects within becoming public.
Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disable
Enable
Tags - optional (0)
You can use bucket tags to track storage costs and organize buckets. Learn more
No tags associated with this bucket.
Add tag


Services
Search
[Alt+S]
Global
VaishnaviGolhar
Default encryption Info
Server-side encryption is automatically applied to new objects stored in this bucket.
Encryption type Info
Server-side encryption with Amazon S3 managed keys (SSE-S3)
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
Disable
Enable
Advanced settings
Object Lock
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. Learn more
Disable
Enable
Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.


Services
Search
[Alt+S]
Global
VaishnaviGolhar
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
Disable
Enable
Advanced settings
Object Lock
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. Learn more
Disable
Enable
Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.
Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.
After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.
Cancel
Create bucket


Services
Search
[Alt+S]
Global
VaishnaviGolhar
Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
Asia Pacific (Mumbai) ap-south-1
Bucket name Info
project1s3phpwebsit
Bucket name must be unique within the global namespace and follow the bucket naming rules. See rules for bucket naming
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix






```
ubuntu@ip-172-31-10-153:/var/www/html$ wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
index.php: Permission denied
ubuntu@ip-172-31-10-153:/var/www/html$ index.php: Permission denied
index.php:: command not found
ubuntu@ip-172-31-10-153:/var/www/html$ ls -ld /var/www/html
drwxr-xr-x 2 root root 4096 Mar 30 18:17 /var/www/html
ubuntu@ip-172-31-10-153:/var/www/html$ sudo chmod 755 /var/www/html
ubuntu@ip-172-31-10-153:/var/www/html$ ls -l /var/www/html
total 0
-rw-r--r-- 1 root root 0 Mar 30 18:17 index.php
ubuntu@ip-172-31-10-153:/var/www/html$ wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
index.php: Permission denied
ubuntu@ip-172-31-10-153:/var/www/html$ -rw-r--r-- 1 root root 0 Mar 30 18:17 index.php
-rw-r--r--: command not found
ubuntu@ip-172-31-10-153:/var/www/html$ sudo chmod 644 index.php
ubuntu@ip-172-31-10-153:/var/www/html$ sudo wget https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
--2024-03-30 18:49:55--  https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
Resolving project1s3phpwebsit.s3.ap-south-1.amazonaws.com (project1s3phpwebsit.s3.ap-south-1.amazonaws.com)... 52.219.156.18, 52.219.156.158, 52.219.158.206, ...
Connecting to project1s3phpwebsit.s3.ap-south-1.amazonaws.com (project1s3phpwebsit.s3.ap-south-1.amazonaws.com)|52.219.156.18|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2135 (2.1K) [binary/octet-stream]
Saving to: 'index.php.1'

index.php.1                          100%[=============================================================================================>]   2.08K  --.-KB/s    in 0s

2024-03-30 18:49:55 (114 MB/s) - 'index.php.1' saved [2135/2135]
```

```
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php  index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ nano  index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php
ubuntu@ip-172-31-10-153:/var/www/html$ rm index.php
rm: remove write-protected regular empty file 'index.php'? y
rm: cannot remove 'index.php': Permission denied
ubuntu@ip-172-31-10-153:/var/www/html$ sudo rm index.php
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ mv index.php1 index.php
mv: cannot stat 'index.php1': No such file or directory
ubuntu@ip-172-31-10-153:/var/www/html$ sudo mv index.php1 index.php
mv: cannot stat 'index.php1': No such file or directory
ubuntu@ip-172-31-10-153:/var/www/html$ mv: cannot stat 'index.php1': No such file or directory
sudo touch index.php
Command 'mv:' not found, did you mean:
  command 'mv' from deb coreutils (8.32-4.1ubuntu1)
  command 'mvn' from deb maven (3.6.3-5)
  command 'mvs' from deb libwww-mediawiki-client-perl (0.31-2.2)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-10-153:/var/www/html$ sudo touch index.php
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php  index.php.1
```

```
ubuntu@ip-172-31-10-153:/var/www/html$ mv: cannot stat 'index.php1': No such file or directory
sudo touch index.php
Command 'mv:' not found, did you mean:
  command 'mv' from deb coreutils (8.32-4.1ubuntu1)
  command 'mvn' from deb maven (3.6.3-5)
  command 'mvs' from deb libwww-mediawiki-client-perl (0.31-2.2)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-10-153:/var/www/html$ sudo touch index.php
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php  index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ sudo mv index.php1 index.php
mv: cannot stat 'index.php1': No such file or directory
ubuntu@ip-172-31-10-153:/var/www/html$  index.php.1mv: cannot stat 'index.php1': No such file or directory
index.php.1mv:: command not found
ubuntu@ip-172-31-10-153:/var/www/html$ mv /path/to/index.php.1 /path/to/index.php
mv: cannot stat '/path/to/index.php.1': No such file or directory
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php  index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ rm index.php.1
rm: remove write-protected regular file 'index.php.1'? sudo rm  index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php  index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ ubuntu@ip-172-31-10-153:/var/www/html$ rm index.php.1
rm: remove write-protected regular file 'index.php.1'? sudo rm  index.php.1
-bash: ubuntu@ip-172-31-10-153:/var/www/html$: No such file or directory
Command 'rm:' not found, did you mean:
  command 'rmt' from deb tar (1.34+dfsg-1ubuntu0.1.22.04.1)
  command 'rmt' from deb dump (0.4b47-3)
  command 'rm' from deb coreutils (8.32-4.1ubuntu1)
  command 'rmm' from deb mailutils-mh (1:3.14-1)
  command 'rmm' from deb mmh (0.4-4)
  command 'rmm' from deb nmh (1.7.1-11)
  command 'rme' from deb pvm-examples (3.4.6-3.2)
  command 'rmf' from deb mailutils-mh (1:3.14-1)
  command 'rmf' from deb mmh (0.4-4)
  command 'rmf' from deb nmh (1.7.1-11)
Try: sudo apt install <deb name>
```

```
Try: sudo apt install <deb name>
ubuntu@ip-172-31-10-153:/var/www/html$ sudo rm index.php.1
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php
ubuntu@ip-172-31-10-153:/var/www/html$ sudo rm index.php
ubuntu@ip-172-31-10-153:/var/www/html$ ls
ubuntu@ip-172-31-10-153:/var/www/html$  wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
index.php: Permission denied
ubuntu@ip-172-31-10-153:/var/www/html$ ls -ld /var/www/html
drwxr-xr-x 2 root root 4096 Mar 30 19:03 /var/www/html
ubuntu@ip-172-31-10-153:/var/www/html$ sudo chmod 644 index.php
chmod: cannot access 'index.php': No such file or directory
ubuntu@ip-172-31-10-153:/var/www/html$ wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
index.php: Permission denied
ubuntu@ip-172-31-10-153:/var/www/html$ ubuntu@ip-172-31-10-153:/var/www/html$ wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
index.php: Permission denied
-bash: ubuntu@ip-172-31-10-153:/var/www/html$: No such file or directory
index.php:: command not found
ubuntu@ip-172-31-10-153:/var/www/html$ sudo wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
--2024-03-30 19:06:31--  https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
Resolving project1s3phpwebsit.s3.ap-south-1.amazonaws.com (project1s3phpwebsit.s3.ap-south-1.amazonaws.com)... 52.219.62.107, 52.219.64.59, 52.219.66.11, ...
Connecting to project1s3phpwebsit.s3.ap-south-1.amazonaws.com (project1s3phpwebsit.s3.ap-south-1.amazonaws.com)|52.219.62.107|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2135 (2.1K) [binary/octet-stream]
Saving to: 'index.php'

index.php                    100%[===========================================================================================>]   2.08K  --.-KB/s    in 0s

2024-03-30 19:06:31 (89.0 MB/s) - 'index.php' saved [2135/2135]

ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php
ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php
ubuntu@ip-172-31-10-153:/var/www/html$
```

5. Update Website Configuration:

Modify the PHP website configuration to connect to the RDS database. Update the hostname, username, password, and database name in the website's configuration file to point to the RDS instance.

```php
$firstname=$_POST['firstname'];
$email=$_POST['email'];
$servername = "database-1.cfqs0oommlil.ap-south-1.rds.amazonaws.com";
$username = "admin";
$password = "Root1234";
$db = "intellipaatdb";
```

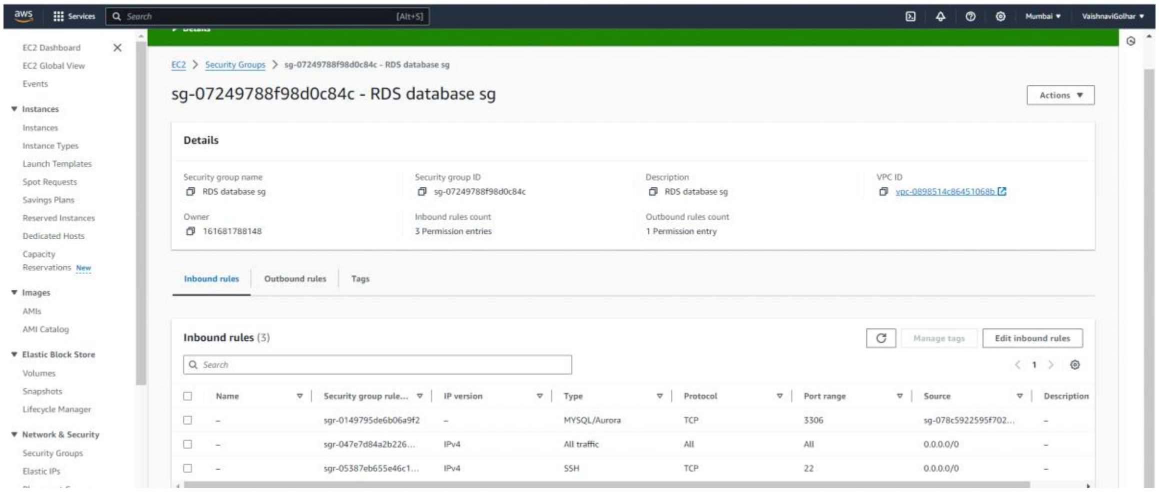

```php
$firstname=$_POST['firstname'];
$email=$_POST['email'];
$servername = "database-1.cfqs0oommlil.ap-south-1.rds.amazonaws.com";
$username = "admin";
$password = "Root1234";
$db = "intellipaatdb";
```


```
ubuntu@ip-172-31-10-153:/var/www/html$ ls
index.php
ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php
ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php
ubuntu@ip-172-31-10-153:/var/www/html$ sudo chmod u+w index.php
ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php
ubuntu@ip-172-31-10-153:/var/www/html$ sudo chown ubuntu:index.php index.php
chown: invalid group: 'ubuntu:index.php'
ubuntu@ip-172-31-10-153:/var/www/html$ sudo chown username:index.php index.php
chown: invalid user: 'username:index.php'
ubuntu@ip-172-31-10-153:/var/www/html$ chown: invalid user: 'username:index.php'
Command 'chown:' not found, did you mean:
  command 'chown' from deb coreutils (8.32-4.1ubuntu1)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-10-153:/var/www/html$ sudo rm index.php
ubuntu@ip-172-31-10-153:/var/www/html$ ls
ubuntu@ip-172-31-10-153:/var/www/html$ sudo wget -O index.php https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
--2024-03-30 19:23:05--  https://project1s3phpwebsit.s3.ap-south-1.amazonaws.com/index.php
Resolving project1s3phpwebsit.s3.ap-south-1.amazonaws.com (project1s3phpwebsit.s3.ap-south-1.amazonaws.com)... 16.12.40.26, 52.219.64.76, 52.219.64.123, ...
Connecting to project1s3phpwebsit.s3.ap-south-1.amazonaws.com (project1s3phpwebsit.s3.ap-south-1.amazonaws.com)|16.12.40.26|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2147 (2.1K) [binary/octet-stream]
Saving to: 'index.php'

index.php                    100%[==============================================================================>]   2.10K  --.-KB/s    in 0s

2024-03-30 19:23:05 (85.2 MB/s) - 'index.php' saved [2147/2147]

ubuntu@ip-172-31-10-153:/var/www/html$ nano index.php
ubuntu@ip-172-31-10-153:/var/www/html$
```

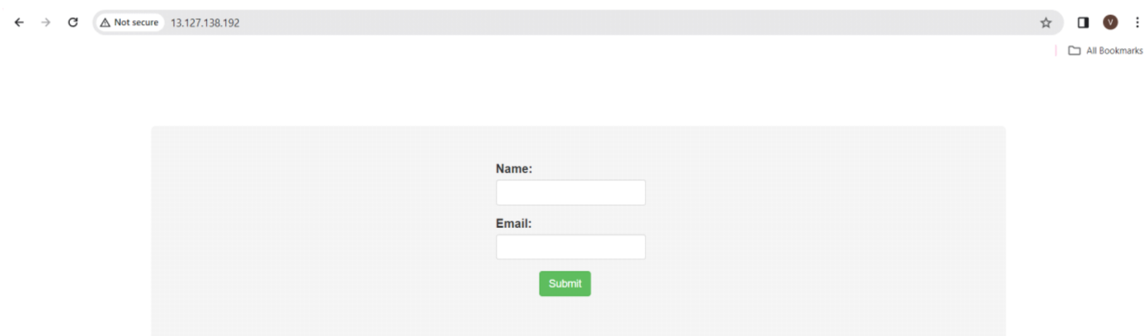
6. Allow Traffic from EC2 to RDS:

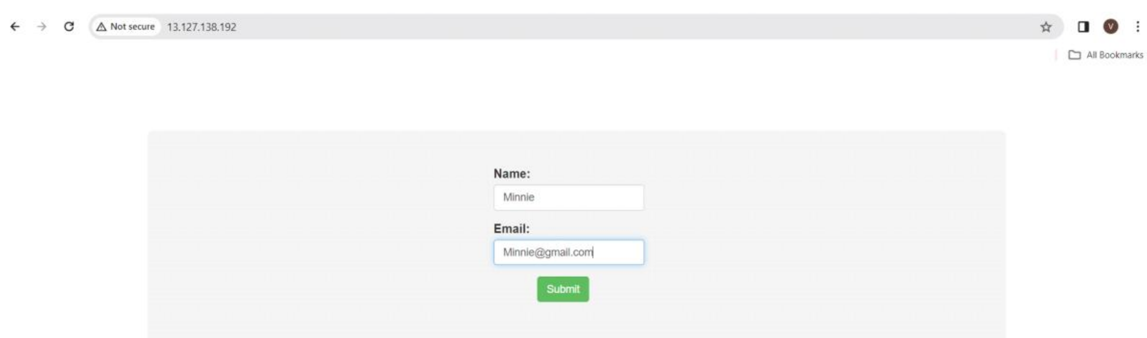
Configure the security group associated with the RDS instance to allow inbound traffic from the security group associated with the EC2 instances hosting the PHP website.

Ensure that the appropriate port (usually 3306 for MySQL) is open in the security group rules.



7. Copy the public address of server and open in new tab

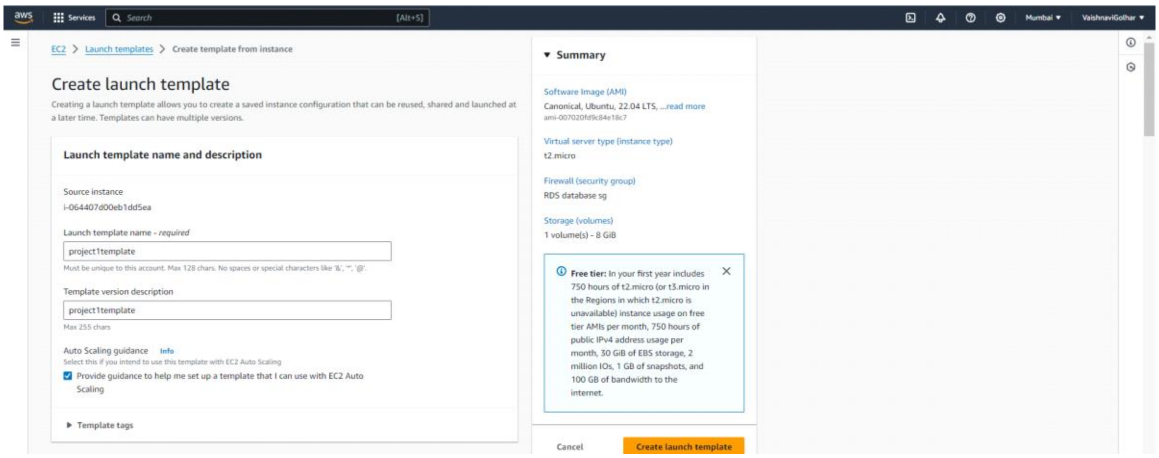


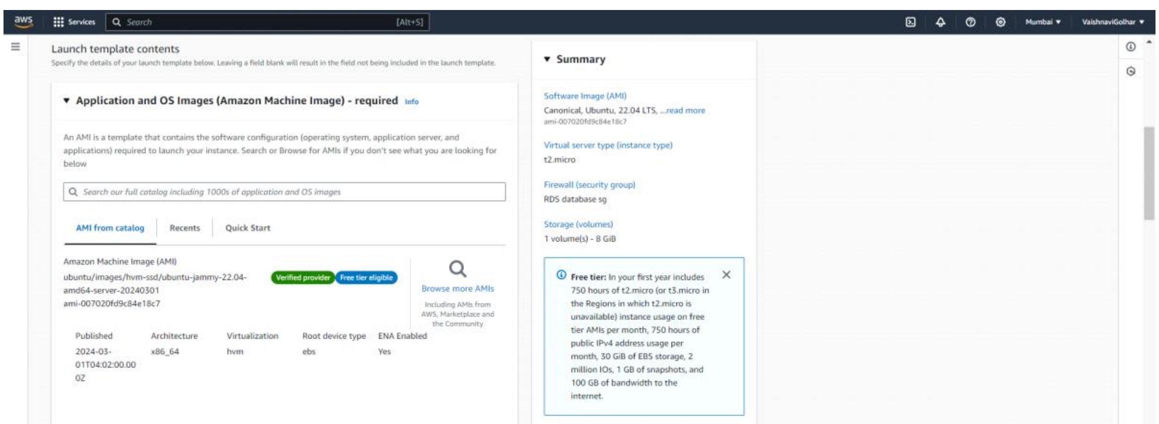


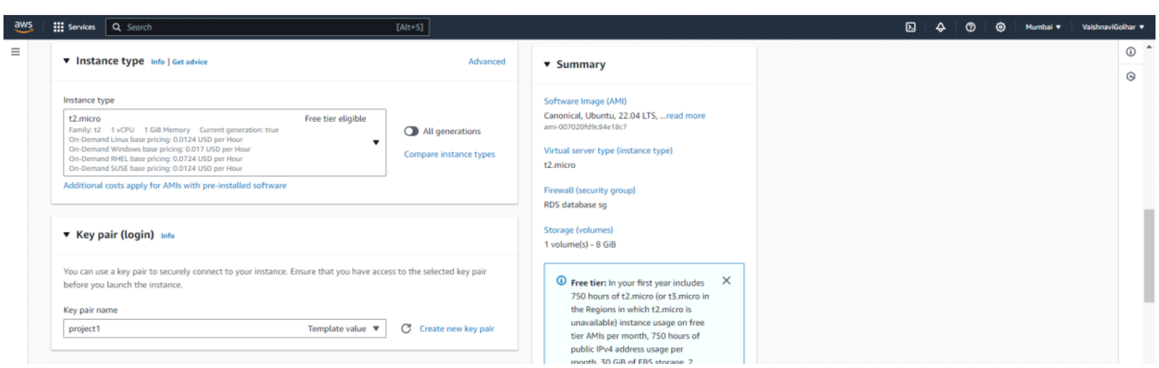
8. Enable Auto Scaling:

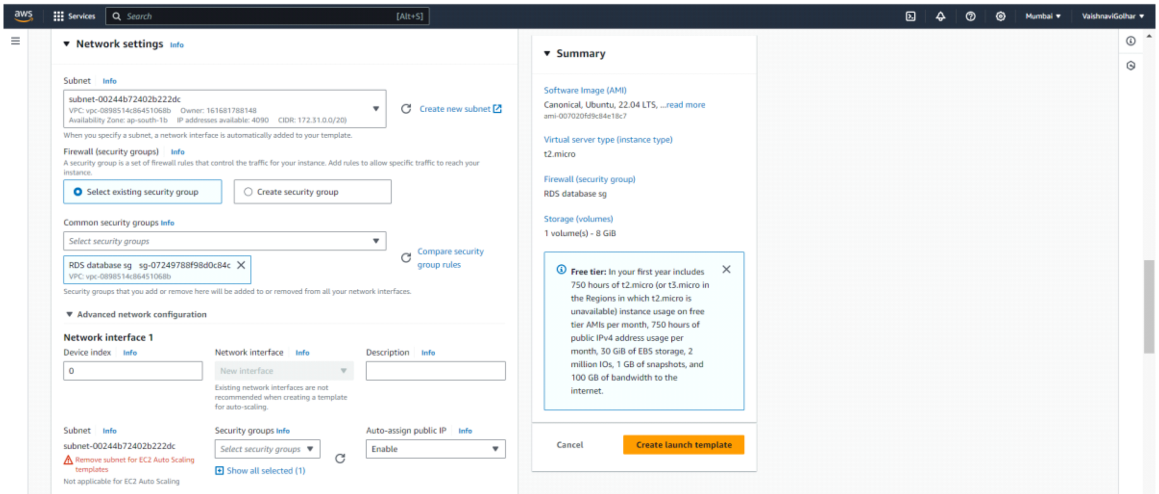
Set up an Auto Scaling Group for your EC2 instances to automatically adjust the number of instances based on demand.

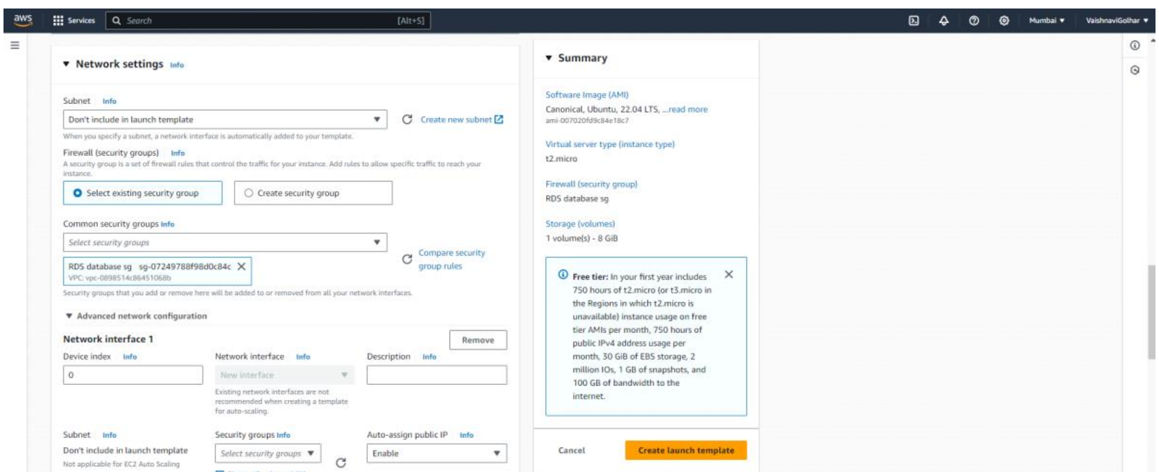
Configure scaling policies based on metrics such as CPU utilization or network traffic to automatically add or remove instances as needed.

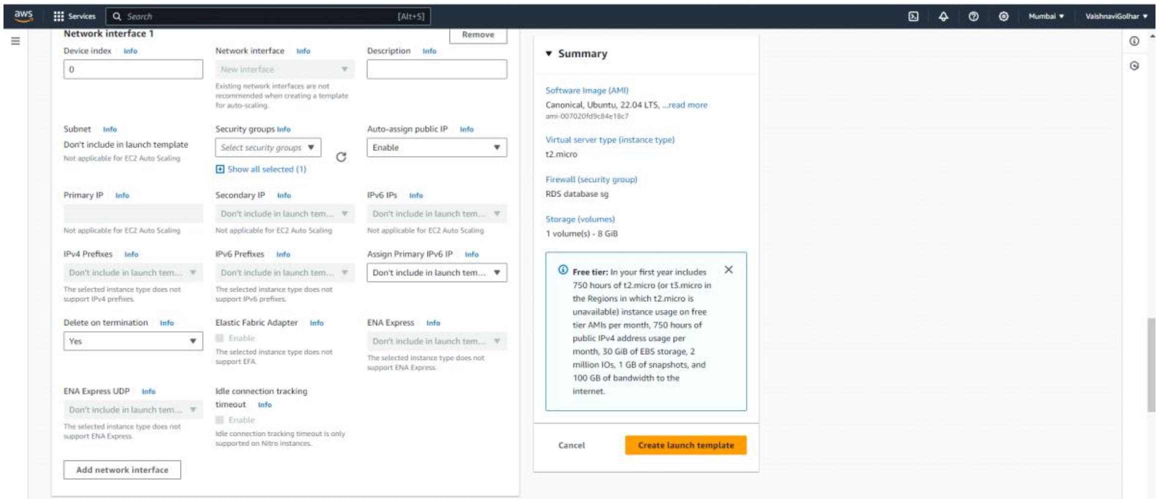


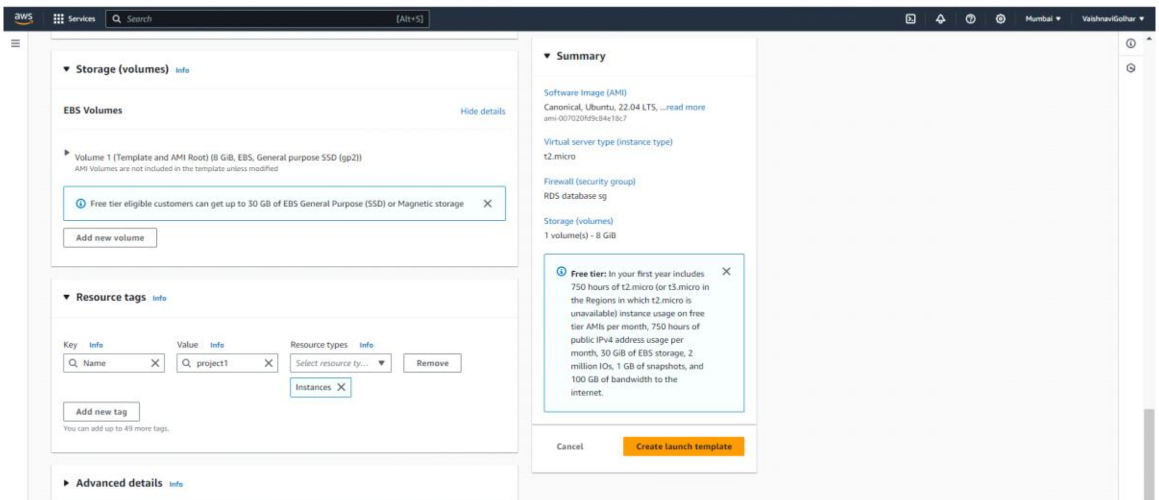


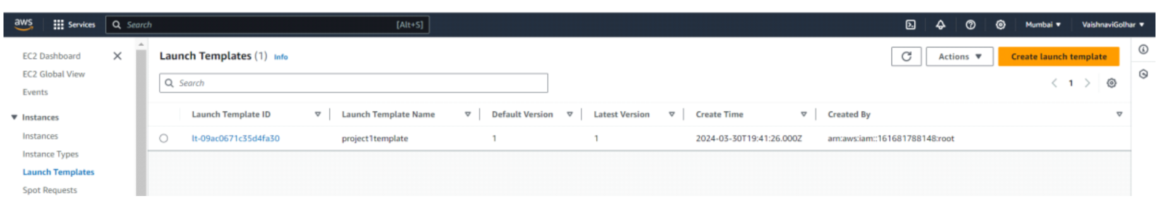


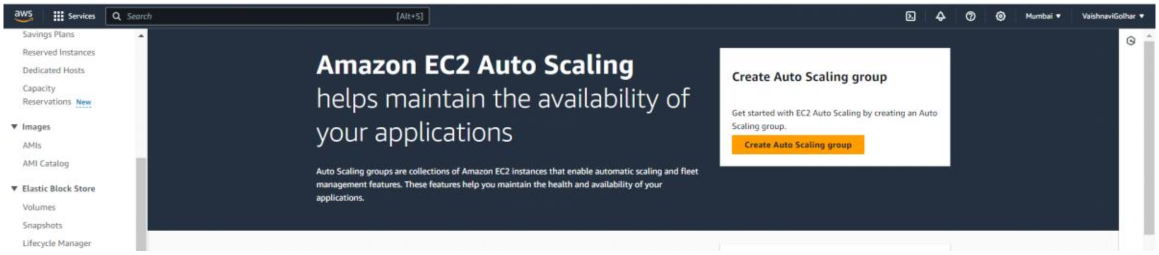

Network settings Info
Subnet Info
subnet-00244b72402b222dc
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Firewall (security groups) Info
Select existing security group
Create security group
Common security groups Info
Select security groups
RDS database sg sg-07249788f98d0c84c
Compare security group rules
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Network interface 1
Device index Info
Network interface Info
Description Info
New interface
Subnet Info
Security groups Info
Auto-assign public IP Info
Enable
Show all selected (1)
Not applicable for EC2 Auto Scaling
Summary
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
RDS database sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.
Cancel
Create launch template

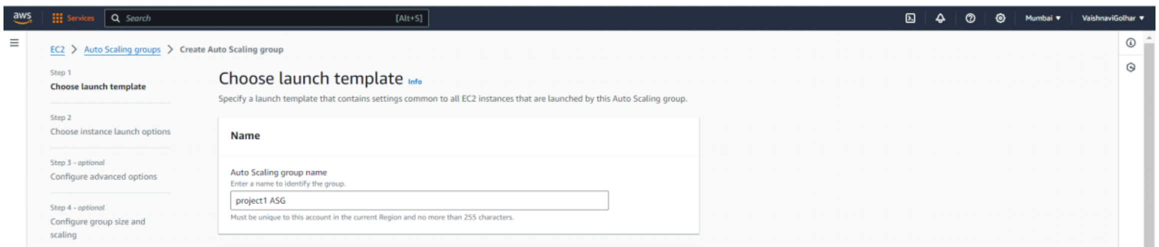

Network settings Info
Subnet Info
Don't include in launch template
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Firewall (security groups) Info
Select existing security group
Create security group
Common security groups Info
Select security groups
RDS database sg sg-07249788f98d0c84c
Compare security group rules
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Network interface 1
Remove
Device index Info
Network interface Info
Description Info
New interface
Subnet Info
Don't include in launch template
Not applicable for EC2 Auto Scaling
Security groups Info
Auto-assign public IP Info
Enable
Show all selected (1)
Summary
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
RDS database sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.
Cancel
Create launch template

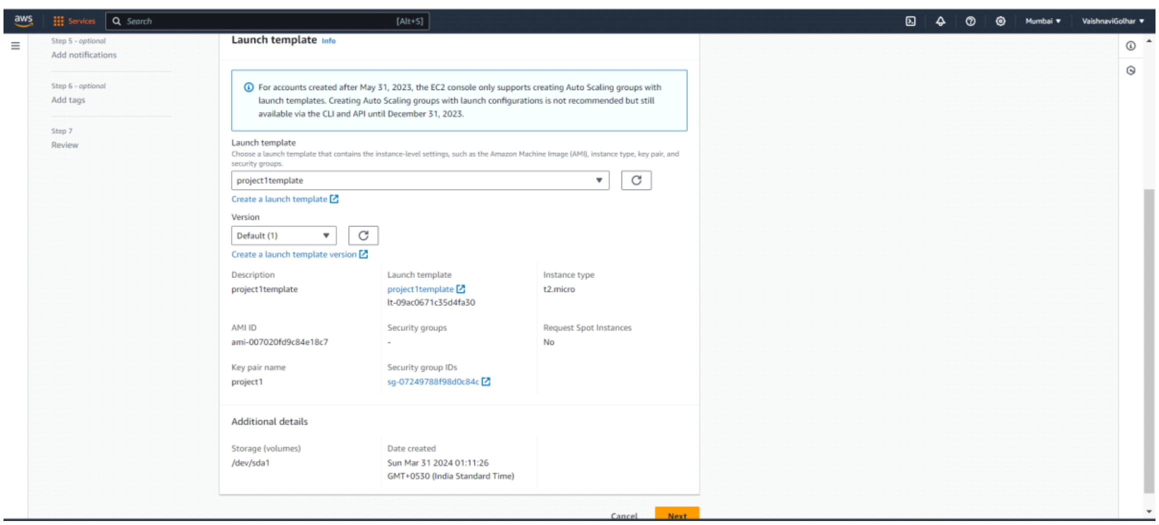

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
Network interface 1
Remove
Device index
Network interface
Description
New interface
Existing network interfaces are not recommended when creating a template for auto-scaling.
Subnet
Don't include in launch template
Not applicable for EC2 Auto Scaling
Security groups
Select security groups
Show all selected (1)
Auto-assign public IP
Enable
Primary IP
Secondary IP
IPv6 IPs
Don't include in launch tem...
Not applicable for EC2 Auto Scaling
IPv4 Prefixes
IPv6 Prefixes
Assign Primary IPv6 IP
The selected instance type does not support IPv4 prefixes.
The selected instance type does not support IPv6 prefixes.
Delete on termination
Yes
Elastic Fabric Adapter
Enable
The selected instance type does not support EFA.
ENA Express
The selected instance type does not support ENA Express.
ENA Express UDP
Idle connection tracking timeout
Idle connection tracking timeout is only supported on Nitro instances.
Add network interface
Summary
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
ami-007020fd9c84e18c7
Virtual server type (instance type)
t2.micro
Firewall (security group)
RDS database sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.
Cancel
Create launch template

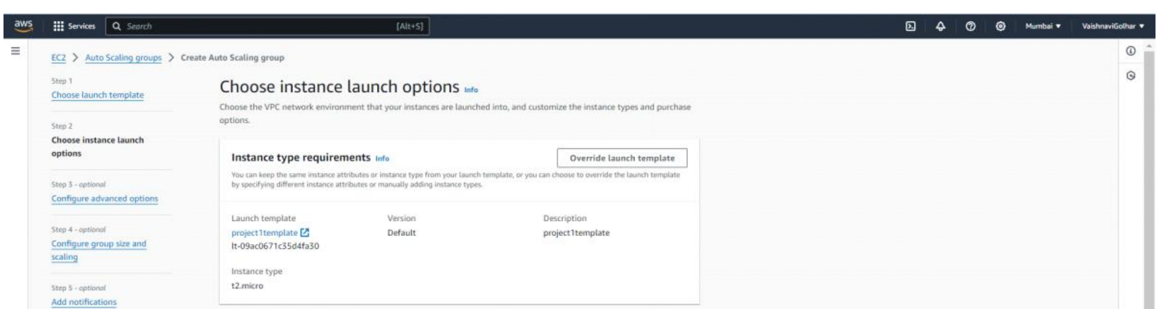

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
Storage (volumes)
EBS Volumes
Hide details
Volume 1 (Template and AMI Root) (8 GiB, EBS, General purpose SSD (gp2))
AMI Volumes are not included in the template unless modified
Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage
Add new volume
Resource tags
Key
Value
Resource types
Name
project1
Select resource ty...
Remove
Instances
Add new tag
You can add up to 49 more tags.
Advanced details
Summary
Software Image (AMI)
Canonical, Ubuntu, 22.04 LTS, ...read more
ami-007020fd9c84e18c7
Virtual server type (instance type)
t2.micro
Firewall (security group)
RDS database sg
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.
Cancel
Create launch template

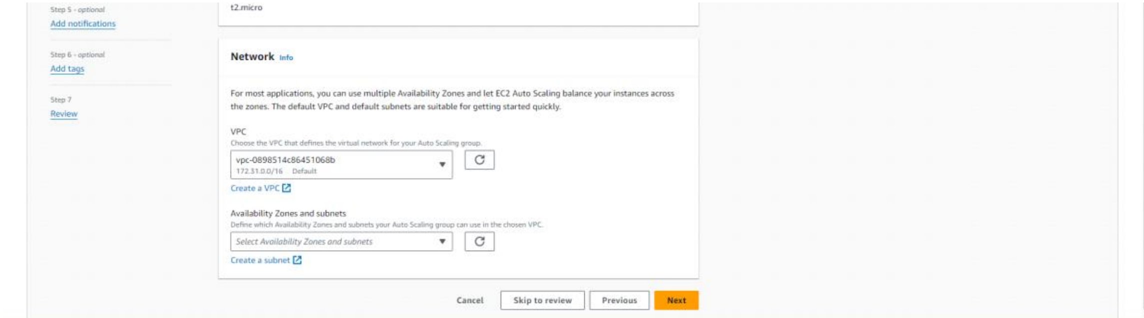

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
EC2 Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Launch Templates (1)
Actions
Create launch template
Search
Launch Template ID
Launch Template Name
Default Version
Latest Version
Create Time
Created By
lt-09ac0671c35d4fa30
project1template
1
1
2024-03-30T19:41:26.000Z
arn:aws:iam::161681788148:root

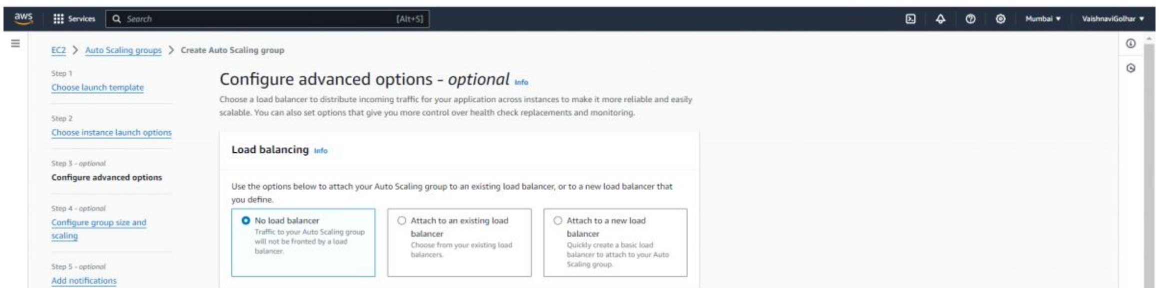

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Amazon EC2 Auto Scaling
helps maintain the availability of your applications
Auto Scaling groups are collections of Amazon EC2 instances that enable automatic scaling and fleet management features. These features help you maintain the health and availability of your applications.
Create Auto Scaling group
Get started with EC2 Auto Scaling by creating an Auto Scaling group.
Create Auto Scaling group

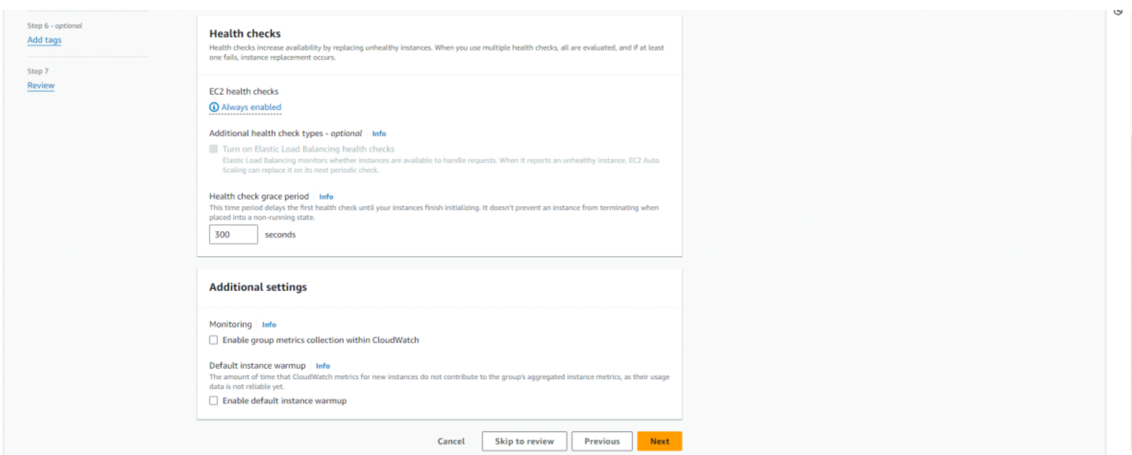

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Configure advanced options
Step 4 - optional
Configure group size and scaling
Choose launch template Info
Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.
Name
Auto Scaling group name
Enter a name to identify the group.
project1 ASG
Must be unique to this account in the current Region and no more than 255 characters.

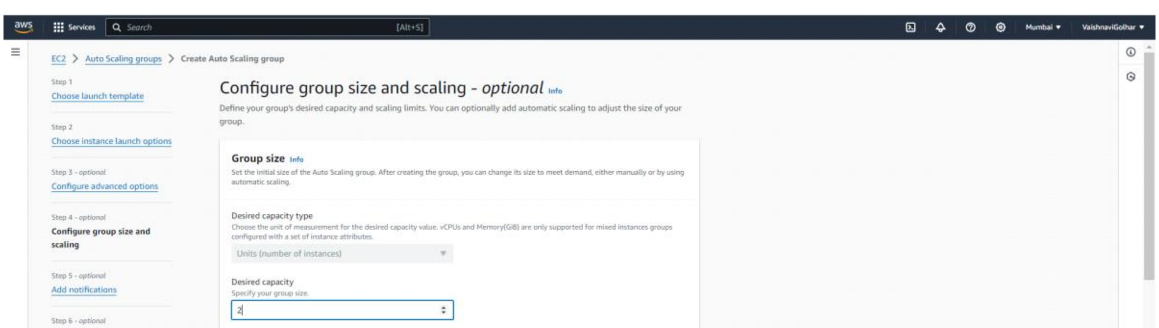

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Launch template Info
For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.
Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
project1template
Create a launch template
Version
Default (1)
Create a launch template version
Description
project1template
Launch template
project1template
lt-09ac0671c35d4fa30
Instance type
t2.micro
AMI ID
ami-007020fd9c84e18c7
Security groups
-
Request Spot Instances
No
Key pair name
project1
Security group IDs
sg-07249788f98d0c84c
Additional details
Storage (volumes)
/dev/sda1
Date created
Sun Mar 31 2024 01:11:26 GMT+0530 (India Standard Time)
Cancel
Next

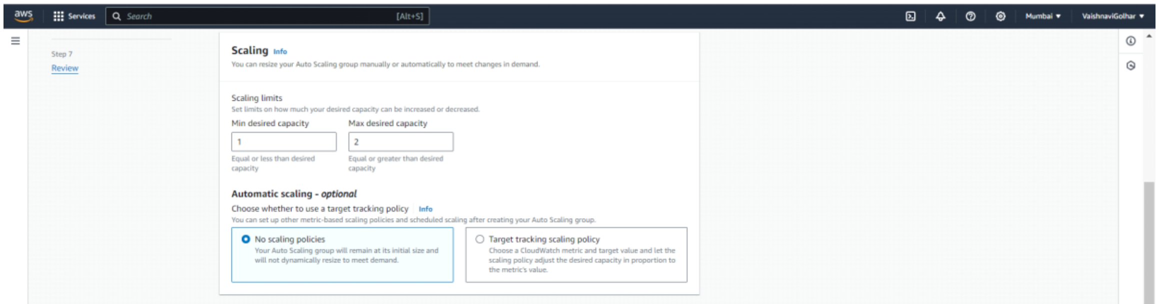

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Configure advanced options
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Choose instance launch options Info
Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.
Instance type requirements Info
Override launch template
You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.
Launch template
project1template
lt-09ac0671c35d4fa30
Version
Default
Description
project1template
Instance type
t2.micro

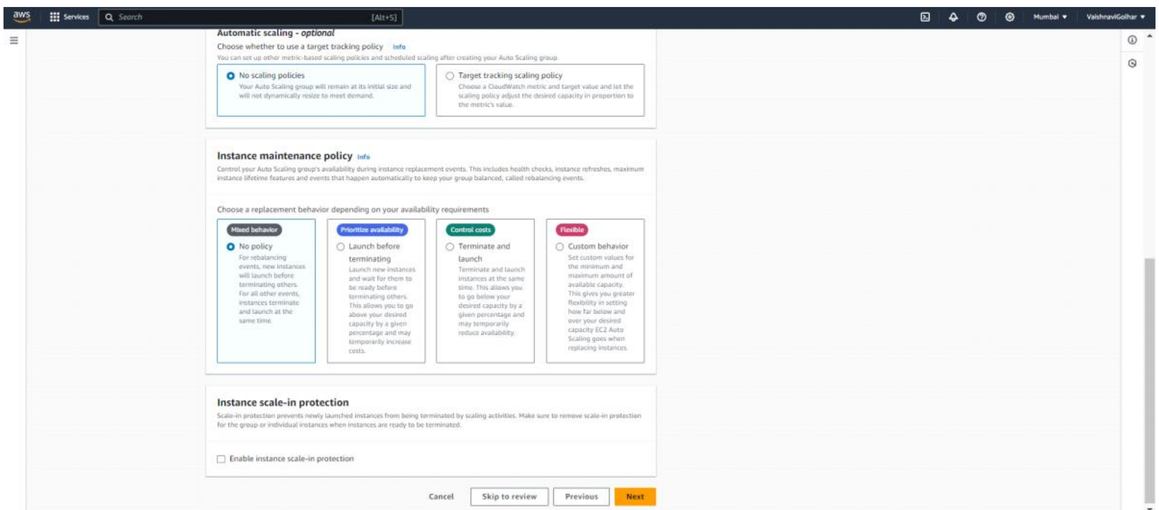

Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
t2.micro
Network Info
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-0898514c86451068b
172.31.0.0/16 Default
Create a VPC
Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
Create a subnet
Cancel
Skip to review
Previous
Next

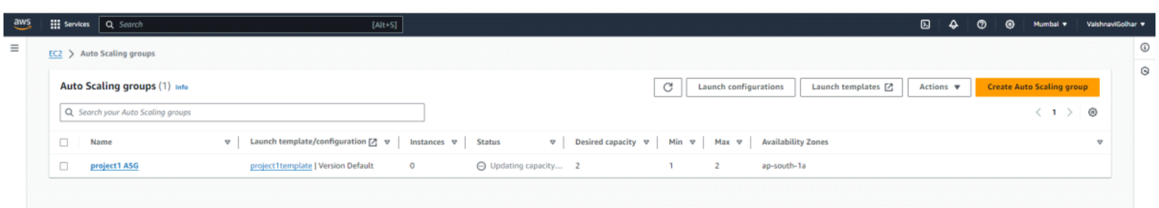

Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Configure advanced options
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Configure advanced options - optional Info
Choose a load balancer to distribute incoming traffic for your application across instances to make it more reliable and easily scalable. You can also set options that give you more control over health check replacements and monitoring.
Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.
No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
Attach to an existing load balancer
Choose from your existing load balancers.
Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

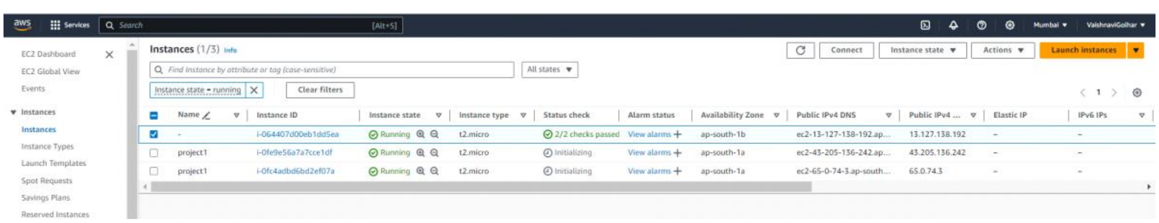

Step 6 - optional
Add tags
Step 7
Review
Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.
EC2 health checks
Always enabled
Additional health check types - optional Info
Turn on Elastic Load Balancing health checks
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.
Health check grace period Info
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.
300
seconds
Additional settings
Monitoring Info
Enable group metrics collection within CloudWatch
Default instance warmup Info
The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.
Enable default instance warmup
Cancel
Skip to review
Previous
Next


Services
Search
[Alt+S]
Mumbai
VaishnaviGolhar
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Configure advanced options
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Configure group size and scaling - optional Info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.
Group size Info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.
Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)
Desired capacity
Specify your group size.
2


Scaling
You can resize your Auto Scaling group manually or automatically to meet changes in demand.
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity
Max desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy
No scaling policies
Target tracking scaling policy


Automatic scaling - optional
No scaling policies
Target tracking scaling policy
Instance maintenance policy
Choose a replacement behavior depending on your availability requirements
No policy
Launch before terminating
Terminate and launch
Custom behavior
Instance scale-in protection
Enable instance scale-in protection
Cancel
Skip to review
Previous
Next


EC2 > Auto Scaling groups
Auto Scaling groups (1)
Launch configurations
Launch templates
Actions
Create Auto Scaling group
project1 ASG
project1template | Version Default
Updating capacity...
ap-south-1a


Instances (1/3)
Connect
Instance state
Actions
Launch instances
Instance state = running
Clear filters
project1
project1
Running
t2.micro
2/2 checks passed
Initializing
View alarms
ap-south-1b
ap-south-1a
13.127.138.192
43.205.136.242
65.0.74.3